Highlights

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

- Knowledge-guided dual-swarm framework for high-dimensional feature selection.
- MI soft prior seeding and adaptive recovery mitigate search stagnation.
- Differential importance memory with exponential decay steers elite refinement.
- Dynamic two-phase scheduling decouples coarse pruning from fine-grained tuning.
- Attains 95.45% internal benchmark CV accuracy with 87.8 features and 3.4x speedup.

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

ARTICLE INFO

Keywords:
Feature selection
Particle swarm optimization
Mutual information
Knowledge-guided elite refinement
Two-phase weight scheduling

ABSTRACT

Feature selection in ultra-high-dimensional and sample-scarce regimes presents severe combinatorial complexity and heightened risk of classifier overfitting. Evolutionary swarm algorithms encounter three structural search degenerations: undirected initialization provokes severe initial search blindness in sparse feature spaces, memoryless uniform perturbation fails to harvest cross-iteration feature feedback, and static objective weighting causes a fundamental phase misalignment across evolutionary stages. To overcome these bottlenecks, this paper proposes Knowledge-Guided Dual-Swarm Evolutionary Learning (KDS-EL), a domain-knowledge-guided feature selection framework established upon a cooperative multi-task dual-swarm PSO backbone. Specifically, mutual-information prior seeding and adaptive recovery substantially alleviate early evaluation waste while preserving swarm diversity. A differential importance memory model with exponential forgetting guides targeted elite refinement under controlled evaluation overhead. Furthermore, dynamic two-phase weight scheduling temporally decouples early dimensional pruning from late classification refinement. Extensive empirical evaluations demonstrate that KDS-EL achieves competitive overall performance under multi-criteria evaluation, yielding an overall mean benchmark-level internal cross-validation accuracy of 95.45% on the shared evaluation surrogate while isolating an ultra-sparse feature subset averaging merely 87.8 active features and achieving a 3.4-fold empirical acceleration over canonical dual-swarm search under the reported implementation environment. Non-parametric Friedman rankings and post-hoc statistical tests corroborate its favorable omnibus ranking across heterogeneous problem domains, demonstrating a balanced trade-off between combinatorial search efficacy, extreme dimensional parsimony, and runtime efficiency.

1. Introduction

High-dimensional data present fundamental challenges to pattern classification across diverse domains, including bioinformatics, transcriptomics, and biomedical diagnostics. These domains frequently involve datasets containing thousands or tens of thousands of continuous features alongside severely restricted sample sizes, creating ultra-high-dimensional and sample-scarce regimes where candidate dimensions $D \gg 10^3$ vastly exceed available observational samples $N_{\text{samp}} \ll 100$. Under such adverse conditions, conventional machine learning algorithms suffer from severe overfitting, prohibitive computational costs, and degraded model interpretability due to the curse of dimensionality [1, 2]. Feature selection has consequently emerged as an indispensable preprocessing technique aimed at identifying a minimal, highly discriminative subset of relevant features while discarding redundant and uninformative dimensions. Nevertheless, discovering an optimal feature subset is an inherently combinatorial problem whose discrete search space scales exponentially as 2^D, rendering exhaustive evaluation NP-hard [3]. Furthermore, univariate feature ranking criteria evaluate dimensions in isolation, fundamentally failing to capture multi-feature synergy and non-linear interactions essential for accurate classification.

Feature selection methods are broadly categorized into filter, wrapper, and embedded paradigms [4, 1]. Filter methods evaluate candidate features based on intrinsic statistical or information-theoretic metrics independently of inductive classifiers, offering high computational speed but frequently overlooking complex multi-feature synergy. Embedded methods incorporate feature selection directly into learning model parameter estimation, providing intermediate computational efficiency; however, their selection performance remains architecturally tied to specific classifiers. Conversely, wrapper methods directly guide feature subset exploration using inductive classifier feedback, naturally capturing non-linear feature interactions and yielding superior classification accuracy, albeit at the expense of intensive computational evaluation. To navigate the combinatorial discrete search space of wrapper feature selection, evolutionary computation has been extensively investigated as a powerful population-based optimization paradigm [5, 3, 27]. By maintaining a diverse population of candidate solutions, evolutionary algorithms explore the discrete combinatorial space without requiring gradient information or restrictive convexity assumptions. Among various evolutionary meta-heuristics, Particle Swarm Optimization (PSO) has demonstrated prominent utility in feature selection tasks due to its conceptual simplicity, rapid information-sharing dynamics, and flexible velocity-position representation [6, 7, 8].

Despite its widespread adoption, canonical PSO encounters three critical structural bottlenecks when navigating ultra-high-dimensional and sample-scarce search spaces. First, standard uniform random initialization scatters particles indiscriminately across the candidate space, whereas optimal feature subsets occupy an ultra-sparse combinatorial subspace with cardinality $d^\star \ll D$. Consequently, a substantial volume of expensive wrapper evaluations is squandered on dense, uninformative candidate masks, severely exacerbating early evaluation latency and sample-scarce overfitting. Second, conventional uniform variation heuristics treat all candidate dimensions equally and operate memorylessly across generations; by discarding the historical feature-level fitness feedback accumulated during previous iterations, the swarm fails to exploit verified discriminative dimensions and degenerates into stochastic random drift. Third, existing evolutionary wrappers enforce static, fixed objective weights to balance classification accuracy against subset cardinality throughout evolution. This static formulation creates a fundamental evolutionary phase misalignment: the search inherently demands aggressive dimensional pruning during early exploration to substantially contract vast noise subspaces, whereas fine-grained decision-boundary tuning is essential during late exploitation to maximize inductive classification accuracy.

Recent evolutionary feature selection studies have investigated specialized strategies to mitigate high-dimensional search degradation, including feature-weight velocity steering in FWPSO [9], adaptive differential evolution weighting in SaWDE [10], surrogate-guided perturbation in IGPSO [11], and multi-criterion evaluation in PSO-CSM [12]. Although these single-objective approaches yield valuable localized improvements, they address initialization, perturbation, and weighting in isolation, lacking a unified closed-loop architecture. Concurrently, multi-task evolutionary optimization has emerged as a promising paradigm for scaling feature selection to massive dimensions by exploring complementary continuous and discrete representations across collaborative subpopulations [13]. Nevertheless, existing multi-task swarm frameworks leave critical structural vulnerabilities unresolved: continuous particles remain uniformly scattered without domain prior guidance, empirical feature-importance accumulation lacks exponential forgetting and elitist verification, and static objective scalarization persists throughout evolution.

To resolve these collective limitations, this paper proposes Knowledge-Guided Dual-Swarm Evolutionary Learning (KDS-EL). Methodologically, KDS-EL establishes an integrated cooperative co-evolutionary system that synergistically couples domain-knowledge seeding, generational memory harvesting, and stage-aware search control across complementary continuous and discrete representations, orchestrating prior injection, differential elite refinement, and dynamic objective scheduling into a closed-loop optimization architecture.

The main contributions of this paper are summarized as follows:

1. *Prior-Guided Sparse Initialization with Adaptive Stagnation Recovery*: A hybrid seeding strategy that embeds continuous mutual-information soft priors into continuous decision variables without irreversible dimensional truncation, preserving full search-space accessibility while integrating Gaussian dispersion and an adaptive stagnation recovery mechanism to mitigate early evaluation waste without sacrificing diversity (Section 3.2).
2. *Knowledge-Guided Elite Refinement (KER)*: A targeted elite refinement strategy that dynamically harvests cross-iteration feature-importance feedback with exponential memory decay, directing elite solutions toward historically effective dimensions within a controlled generational evaluation overhead of $k_e \ll N$ candidates (Section 3.3).

Table 1: Methodological taxonomy of investigated evolutionary feature selection algorithms.

		Knowledge Engineering			Search Control		
Algorithm	Paradigm	MI Prior Seeding	Online Memory	Memory Decay	Elite Refinement	Dynamic Weighting	Evaluation Budget
BPSO [8]	SO-SI	×	×	×	×	×	N
FWPSO [9]	SO-SI	○	×	×	×	×	N
SaWDE [10]	SO-DE	×	×	×	×	○	N
IGPSO† [11]	SO-SI	○	×	×	×	×	N
PSO-CSM [12]	SO-SI	○	×	×	×	×	N
DENCA-SO‡ [14]	SO-DE	○	×	×	×	×	N
bIRIME [15]	SO-SI	×	×	×	○	×	$\leq 4N$
BinCOA [16]	SO-SI	×	×	×	○	×	N
MEL [13]	MT-SI	×	✓	×	×	×	N
KDS-EL (Proposed)	**MT-SI**	✓	✓	✓	✓	✓	$\mathbf{N + k_e}$

Symbols: ✓ Supported; ○ Partial; × Absent. **Paradigms:** SO: Single-Objective; MT: Multi-Task; SI: Swarm Intelligence; DE: Differential Evolution. †IGPSO designates the sample-adapted ANOVA permutation contrast variant detailed in Section 4.1. ‡DENCA-SO utilizes an algebraic filter surrogate instead of wrapper evaluations.

3. *Two-Phase Weight Scheduling (TPS)*: A stage-aware objective scheduling mechanism that temporally decouples early aggressive feature pruning from late-stage accuracy refinement, overcoming the rigidity of static weighting (Section 3.4).
4. *Comprehensive Empirical Benchmark Validation*: Extensive evaluations across 20 high-dimensional benchmark datasets spanning 500 to 54,675 features against nine representative competitor algorithms and intermediate configurations, rigorously validated through non-parametric statistical testing, component contribution analysis, and parametric sensitivity investigations (Section 4).

The remainder of this paper is structured as follows: Section 2 reviews related feature selection paradigms and background concepts; Section 3 details the proposed KDS-EL framework and its constituent mechanisms; Section 4 presents comprehensive experimental evaluations and empirical analysis; and Section 5 concludes the paper with prospective research trajectories.

2. Related Work and Background

2.1. Problem formulation and search paradigms in feature selection

The essence of feature selection lies in combinatorial optimization, aiming to identify an optimal, minimal subset of informative features from a high-dimensional feature space to maximize classification performance while discarding redundant and uninformative dimensions [2, 1]. Formally, let $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$ denote a supervised training dataset comprising N_{samp} instances, where $\mathbf{f}_i \in \mathbb{R}^D$ represents the full feature vector in the original D-dimensional space and $y_i \in \{1, 2, \ldots, C\}$ designates the discrete class label across C categories. In discrete binary representations, a candidate feature subset is formulated as a binary decision indicator vector $\mathbf{x} \in \{0, 1\}^D$, defined as:

$$x_j = \begin{cases} 1, & \text{if feature } f_j \text{ is selected,} \\ 0, & \text{if feature } f_j \text{ is discarded.} \end{cases} \tag{1}$$

The primary objective of feature selection is to simultaneously minimize the classification error rate and the cardinality of the selected feature subset. In evolutionary wrapper feature selection, this multi-criterion goal is mathematically formalized as minimizing a scalarized single-objective fitness function $f(\mathbf{x})$ [17, 18]:

$$\min_{\mathbf{x}\in\{0,1\}^D} f(\mathbf{x}) = w_a \cdot \text{ErrorRate}(\mathbf{x}) + w_f \cdot \left(\frac{\|\mathbf{x}\|_0}{D}\right), \tag{2}$$

subject to $\mathbf{x} \in \{0, 1\}^D$ and $w_a + w_f = 1$, where ErrorRate($\mathbf{x}$) represents the cross-validation classification error evaluated on the masked feature subset $\{\mathbf{f}_i \odot \mathbf{x}\}$ using an inductive k-nearest neighbors (k-NN) learner, with $\odot$ denoting the Hadamard element-wise product. The term $\|\mathbf{x}\|_0$ denotes the number of active features, $\|\mathbf{x}\|_0/D$ represents the feature selection ratio, and $w_a, w_f \in [0, 1]$ serve as trade-off coefficients balancing classification accuracy against subset compactness. While multi-objective algorithms approximate a Pareto front of trade-off solutions, practical high-throughput screening and clinical diagnosis predominantly favor scalarized wrapper formulations [4, 17]. Scalarized optimization yields an unambiguous, single deployable feature signature conforming to predefined operational preferences, avoiding the post-hoc decision ambiguity inherent in non-dominated archives. Because exhaustive evaluation across the 2^D combinatorial space is NP-hard when $D \gg 10^3$, wrapper meta-heuristics directly utilize classifier feedback to capture non-linear feature interactions without restrictive convexity assumptions.

To navigate this combinatorial space, evolutionary computation paradigms, including genetic algorithms [24], particle swarm optimization [25], hybrid evolutionary optimization [26], and swarm intelligence [5, 3, 28], serve as dominant search heuristics. Advanced single-objective strategies have introduced specialized mechanisms to combat high dimensionality, including adaptive weighting in SaWDE [10], offline feature weighting in FWPSO [9], permutation-contrast guidance in IGPSO [11], and multi-criterion restriction in PSO-CSM [12]. Nevertheless, these single-objective wrappers treat feature dimensions uniformly during initialization, discard cross-generational fitness feedback, and enforce static objective trade-offs, precipitating severe search degradation in ultra-high-dimensional spaces.

2.2. Evolutionary swarm intelligence and discrete feature selection dynamics

Standard particle swarm optimization, originally introduced by Kennedy and Eberhart [6], is a population-based stochastic meta-heuristic inspired by the collective foraging behaviors of bird flocks and fish schools. A swarm consists of N candidate particles navigating a D-dimensional continuous solution space $\mathbb{R}^D$. At iteration t, each particle i is characterized by a continuous position vector $\mathbf{p}_i^{(t)} \in \mathbb{R}^D$ and an associated velocity vector $\mathbf{v}_i^{(t)} \in \mathbb{R}^D$, steered dynamically by its personal historical best position $\mathbf{p}_{\text{best},i} \in \mathbb{R}^D$ and the swarm's global best position $\mathbf{g}_{\text{best}} \in \mathbb{R}^D$. The velocity and position updates of particle i at iteration $t + 1$ are formulated as follows [7]:

$$\mathbf{v}_i^{(t+1)} = \omega \mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},i} - \mathbf{p}_i^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best}} - \mathbf{p}_i^{(t)}), \tag{3}$$

$$\mathbf{p}_i^{(t+1)} = \mathbf{p}_i^{(t)} + \mathbf{v}_i^{(t+1)}, \tag{4}$$

where $\omega \in [0, 1]$ denotes the inertia weight controlling momentum retention, $c_1, c_2 > 0$ are the cognitive and social acceleration coefficients governing the attraction toward individual and collective best experiences, and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are independent uniform random factors providing stochastic exploration capability.

To adapt continuous PSO to the combinatorial nature of binary feature selection where candidate solutions are represented as binary indicator masks $\mathbf{x}_i \in \{0, 1\}^D$, Kennedy and Eberhart [8] proposed Binary PSO (BPSO). In BPSO, continuous velocities are mapped into selection probabilities via a Sigmoid transfer function:

$$S(v_{i,j}^{(t+1)}) = \frac{1}{1 + \exp(-v_{i,j}^{(t+1)})}, \tag{5}$$

and the corresponding binary decision variable $x_{i,j}^{(t+1)}$ is sampled probabilistically as:

$$x_{i,j}^{(t+1)} = \begin{cases} 1, & \text{if } r_{i,j} < S(v_{i,j}^{(t+1)}), \\ 0, & \text{otherwise,} \end{cases} \tag{6}$$

where $r_{i,j} \sim \mathcal{U}(0, 1)$ is an independent uniform random threshold. Although BPSO exhibits fast convergence on low-dimensional benchmarks,

it encounters severe exploration-exploitation dilemmas when applied to ultra-high-dimensional biomedical domains where $D \gg 10^3$. Because the Sigmoid function satisfies $S(0) = 0.5$, continuous velocities hovering near zero inevitably yield an expected active feature density of approximately $D/2$. In typical biomedical datasets containing $D \geq 10{,}000$ dimensions, this stochastic mapping generates candidate subsets with more than 5,000 active features, heavily overloading wrapper classifier evaluations with $O(N_{\text{samp}}^2 d_{\text{selected}})$ complexity and triggering severe overfitting under small sample sizes where $N_{\text{samp}} \ll 100$. Furthermore, continuous stochastic sampling introduces high generational variance: even when velocity vectors stabilize, candidate feature masks continue to jitter stochastically, causing particle trajectories to degrade into memoryless random walks across the combinatorial space.

To mitigate the random drift of standard binarization, diverse single-objective meta-heuristics have been introduced, as systematically categorized in Table 1. Evolutionary filter approaches, such as DENCA-SO [14], optimize algebraic surrogate costs to accelerate execution; however, being decoupled from inductive classification margins, they exhibit elevated feature redundancy. Concurrently, advanced discrete optimizers, including bIRIME [15] and BinCOA [16], deploy composite operators to enhance discrete exploration, but scaling trial vector evaluations per agent severely inflates computational overhead on large-sample benchmarks (Section 4.2.6). Crucially, without proactive sparse guidance, stochastic binarization traps these algorithms near the uninformative 50% dimensionality attractor. Furthermore, wrapper methods employing static scalarization, including SaWDE [10], FWPSO [9], and PSO-CSM [12], enforce fixed trade-off weights throughout evolution, precluding stage-aware adaptation between early dimensional pruning and late accuracy refinement.

2.3. Multi-task evolutionary learning and dual-swarm search

Evolutionary Multi-Task Optimization (EMTO) solves multiple distinct optimization tasks simultaneously by facilitating inter-task knowledge transfer across unified representations [29]. In feature selection, EMTO frameworks divide the population into collaborative subpopulations exploring complementary search subspaces [30, 31, 32, 33, 34]. Notably, Wang et al. [13] established a dual-subpopulation cooperative multi-task framework where a population of N particles is partitioned into collaborative subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$, each comprising $N/2$ individuals operating on complementary representations. Specifically, *Subpopulation 1* executes continuous swarm exploration in $[0,1]^D$, updating particle velocities and positions through PSO dynamics augmented with an inter-task knowledge transfer channel directing particles toward the historical global best position $\mathbf{g}_{\text{best},2} \in \{0,1\}^D$ of Subpopulation 2:

$$\begin{aligned}\mathbf{v}_{1,i}^{(t+1)} = \omega \mathbf{v}_{1,i}^{(t)} &+ c_1 \mathbf{r}_1 \odot (\mathbf{p}_{\text{best},1,i} - \mathbf{x}_{1,i}^{(t)}) \\ &+ c_2 \mathbf{r}_2 \odot (\mathbf{g}_{\text{best},1} - \mathbf{x}_{1,i}^{(t)}) + c_3 \mathbf{r}_3 \odot (\mathbf{g}_{\text{best},2} - \mathbf{x}_{1,i}^{(t)}),\end{aligned} \tag{7}$$

where $\mathbf{x}_{1,i}^{(t)} \in [0,1]^D$ and $\mathbf{v}_{1,i}^{(t)} \in [v_{\min}, v_{\max}]^D$ denote the continuous position and velocity of particle i, $\mathbf{p}_{\text{best},1,i}$ and $\mathbf{g}_{\text{best},1}$ represent personal and local bests, $\omega \in [0,1]$ is the inertia weight, $c_1, c_2, c_3 > 0$ are acceleration coefficients, $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3 \sim \mathcal{U}(0,1)^D$, and $\odot$ denotes the Hadamard product. Continuous positions are binarized into candidate selection masks via transfer mapping. Concurrently, *Subpopulation 2* performs discrete sampling from a shared importance vector $\mathbf{W}_{\text{freq}} \in \mathbb{R}^D$ accumulating historical feature selection frequencies across candidate solutions. Both subpopulations evaluate candidate subsets via a shared scalar fitness function similar to Eq. (2) with fixed static weights $\alpha, \beta \in [0,1]$ satisfying $\alpha + \beta = 1$.

While multi-task swarm learning confirms the efficacy of continuous-discrete knowledge transfer, foundational dual-swarm architectures encounter critical vulnerabilities in ultra-high-dimensional regimes: unguided initialization induces early evaluation latency, raw frequency accumulation lacks memory forgetting and elite verification, and static objective scalarization impedes stage-specific convergence. These unresolved structural challenges motivate the closed-loop design of KDS-EL.

2.4. Knowledge-guided search and taxonomic positioning

Integrating domain knowledge into meta-heuristics effectively mitigates the blindness of black-box combinatorial search [35, 36, 18]. In high-dimensional feature selection, Mutual Information (MI), rooted in Shannon's information theory [37], quantifies non-linear dependency between continuous feature variables $f_j \in \mathbb{R}$ and discrete class labels $Y \in \{1, \ldots, C\}$:

$$\begin{aligned}I(f_j; Y) &= H(f_j) - H(f_j \mid Y) \\ &= \sum_{c=1}^{C} P(Y=c) \int_{\mathbb{R}} p(f_j \mid Y=c) \log_2 \frac{p(f_j \mid Y=c)}{p(f_j)} \, df_j,\end{aligned} \tag{8}$$

where $P(Y=c)$ is the class prior probability, while $p(f_j)$ and $p(f_j \mid Y=c)$ denote the marginal and conditional density functions of continuous feature f_j. To avoid bin-width sensitivity in empirical applications with unknown distributions, non-parametric nearest-neighbor estimators, specifically the Kraskov–Stögbauer–Grassberger (KSG) estimator [38] extended by Ross [39], directly evaluate non-linear dependencies. Crucially, in ultra-high-dimensional and sample-scarce regimes where candidate dimensions $D \gg 10^3$ vastly outnumber observations $N_{\text{samp}} \ll 100$, continuous estimators inevitably incur finite-sample estimation variance, producing ranking noise. Conventional filter methods [19, 18] evaluate mutual information in $O(DN_{\text{samp}} + D \log D)$ time but enforce rigid hard truncation, retaining only the top-k ranked features. This hard cut-off fundamentally ignores multi-feature synergy: weakly individual features that are collectively informative when combined are permanently discarded, while false positives elevated by finite-sample noise are irreversibly locked into the candidate pool.

Existing knowledge-guided evolutionary methods primarily adopt one-shot offline injection via either static subspace pre-selection [36, 18] or static velocity biasing [9]. While beneficial in narrowing early search, one-shot injection lacks online adaptability: the static prior remains frozen and cannot assimilate iterative trial-and-error feedback from inductive classification margins. Conversely, online surrogate-assisted approaches, including neural network surrogates and Gaussian processes [40], attempt dynamic importance learning but incur prohibitive retraining latencies and suffer severe model misspecification under sample-scarce regimes where observations $N_{\text{samp}} \ll 100$ cannot constrain high-capacity surrogates.

To resolve the limitations of rigid static priors and heavyweight surrogates, KDS-EL unifies offline information-theoretic priors, online differential memory harvesting, and stage-aware objective scheduling into a closed-loop evolutionary architecture. Table 1 systematically summarizes the methodological taxonomy and structural comparison between KDS-EL and all nine competitor feature selection algorithms investigated in this study across seven orthogonal technical dimensions.

3. The Proposed Method

3.1. The framework of KDS-EL

The macro data-flow architecture and problem mapping of KDS-EL for high-dimensional feature selection are systematically illustrated in Fig. 1. Given a high-dimensional dataset $\mathcal{D}$ characterized by severe sample scarcity, the search pipeline begins by extracting filter-level mutual information to construct a continuous prior relevance landscape $\mathbf{m} \in [0,1]^D$. An initial population of continuous positions $\mathbf{p}_i \in [0,1]^D$ is generated via hybrid prior seeding and deterministically mapped into sparse binary candidate masks $\mathbf{x}_i \in \{0,1\}^D$ via Heaviside thresholding. Throughout optimization, these candidate subsets are transferred to the KDS-EL optimization engine, which coordinates collaborative dual-swarm co-evolution between subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$, dynamic Two-Phase Weight Scheduling, wrapper classification evaluations, cumulative feature-importance harvesting into memory vector $\mathbf{W} \in \mathbb{R}^D$, and targeted elite refinement in closed-loop synergy. Upon satisfying the termination criterion, the framework isolates the optimal compact feature subset $\mathbf{x}^\star$ with cardinality $d^\star = \|\mathbf{x}^\star\|_0 \ll D$ alongside its cross-validated classification accuracy.

The execution logic and operational flowchart across evolutionary generations are depicted in Fig. 2. In KDS-EL, candidate solutions operate under a continuous-discrete dual representation: continuous coordinates reside in the unit hypercube $\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_N]^T \in [0,1]^{N \times D}$, while the evaluated feature subsets across D dimensions are formally represented as an $N \times D$ binary decision matrix $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix}\mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_N\end{bmatrix} = \begin{bmatrix}x_{1,1} & x_{1,2} & \cdots & x_{1,D} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1} & x_{N,2} & \cdots & x_{N,D}\end{bmatrix} \in \{0,1\}^{N \times D}, \tag{9}$$

where the i-th row vector $\mathbf{x}_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,D}] \in \{0,1\}^D$ denotes an individual candidate solution, with $x_{i,j} = 1$ indicating that feature f_j is selected and $x_{i,j} = 0$ indicating exclusion. The mapping from continuous particle position $\mathbf{p}_i = [p_{i,1}, \ldots, p_{i,D}] \in [0,1]^D$ to binary decision mask $\mathbf{x}_i$ is governed by a deterministic threshold condition $x_{i,j} = \mathbb{I}(p_{i,j} \geq \theta)$, where $\mathbb{I}(\cdot)$ denotes the indicator function and $\theta \in (0.5, 1)$ represents the binarization cut-off threshold.

To balance global exploration and local exploitation in sparse high-dimensional spaces, the swarm is partitioned into two collaborative subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$, each comprising $N/2$ particles, leveraging the

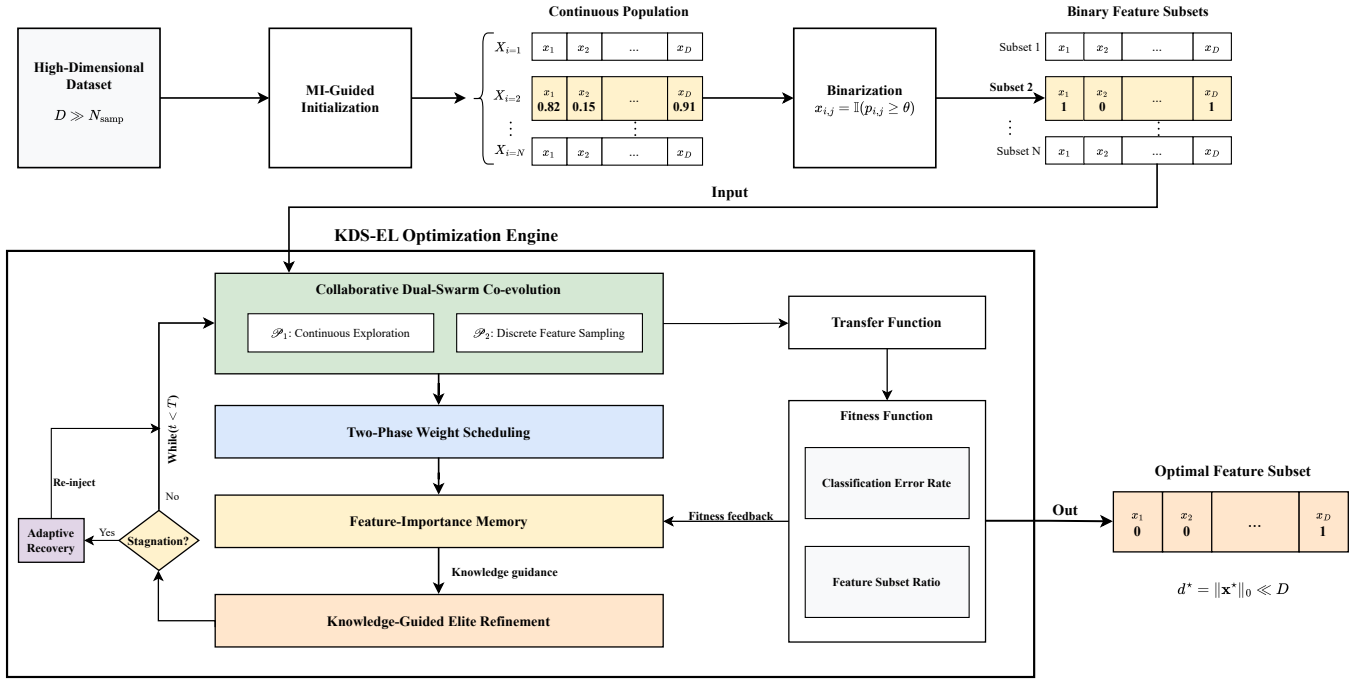


Figure 1: Overall dataflow architecture of the KDS-EL evolutionary framework.

collaborative continuous-discrete multi-task learning architecture established in the original MEL algorithm [13]. As delineated in the upper sub-panel of Fig. 2, Subpopulation $\mathcal{P}_1$ functions as a continuous explorer whose velocities and positions are updated with an inter-task knowledge transfer channel guided by the global best solution $\mathbf{g}_{\text{best},2}$ of $\mathcal{P}_2$:

$$v_{i,j}^{(t)} = \text{clip}\Big(wv_{i,j}^{(t-1)} + c_1 r_{1,j}(p_{\text{best},i,j} - p_{i,j}^{(t-1)}) + c_2 r_{2,j}(g_{\text{best},j} - p_{i,j}^{(t-1)}) + c_3 r_{3,j}(g_{\text{best},2,j} - p_{i,j}^{(t-1)}),\ -v_{\max},\ v_{\max}\Big), \quad (10)$$

$$p_{i,j}^{(t)} = \text{clip}\left(p_{i,j}^{(t-1)} + v_{i,j}^{(t)},\ 0,\ 1\right), \quad x_{i,j}^{(t)} = \mathbb{I}(p_{i,j}^{(t)} \geq \theta),$$

where w is inertia weight, c_1, c_2, c_3 are acceleration coefficients, and $r_{1,j}, r_{2,j}, r_{3,j} \sim \mathcal{U}(0,1)$. Concurrently, Subpopulation $\mathcal{P}_2$ acts as a discrete exploiter, synthesizing candidate masks through independent coordinate-wise Bernoulli sampling $x_{i,j}^{(t)} \sim \text{Bernoulli}\left(\pi_j^{(t)}\right)$, where the dynamic selection probability $\pi_j^{(t)} \in [0,1]$ is derived from the non-negative truncation of the cumulative importance vector $\mathbf{W} \in \mathbb{R}^D$:

$$\pi_j^{(t)} = \begin{cases} \dfrac{\max(W_j, 0)}{\sum_{l=1}^{D} \max(W_l, 0)}, & \text{if } \sum_{l=1}^{D} \max(W_l, 0) > 0, \\ \dfrac{1}{2}, & \text{otherwise.} \end{cases} \quad (11)$$

Because $\sum_{j=1}^{D} \pi_j^{(t)} = 1$ under positive feature credit, this sum-normalized sampling inherited from the original MEL framework operates as an ultra-sparse probe with expected cardinality $\mathbb{E}[\|\mathbf{x}_i\|_0] = 1$. Sub-minimal candidates with $\|\mathbf{x}_i\|_0 < d_{\min}$ are intercepted by the exterior penalty barrier (27) to enforce the minimum cardinality constraint while preserving credit feedback, whereas feasible subsets are collaboratively discovered by the continuous explorer $\mathcal{P}_1$ and refined by the KER strategy. While KDS-EL preserves the dual-swarm multi-task division of (10) and (11) from the original MEL algorithm [13], it fundamentally resolves the search amnesia, unguided initialization, and late-stage stagnation vulnerabilities of the foundational framework by integrating prior-guided sparse seeding, adaptive stagnation recovery, dynamic two-phase objective scheduling, exponential memory forgetting on $\mathbf{W}$, and knowledge-guided elite refinement into a closed loop.

As systematically illustrated in the operational flowchart of Fig. 2, KDS-EL establishes an orthogonal spatio-temporal hierarchy: collaborative dual-swarm co-evolution governs spatial exploration across complementary continuous and discrete domains, while the dynamic TPS strategy orchestrates temporal search progression across decoupled pruning and refinement phases. In closed-loop synergy, prior-guided sparse seeding, differential importance accumulation, and knowledge-guided elite refinement systematically prune redundant dimensions while progressively refining compact, highly discriminative feature subsets.

3.2. Population initialization and adaptive recovery

In ultra-high-dimensional wrapper feature selection, standard uniform random initialization samples continuous particle coordinates uniformly across the unit hypercube, activating each candidate feature with an expected probability of 0.50 and collapsing candidate subsets into the dense dimensionality attractor characterized by cardinality $d \approx D/2$. In high-dimensional regimes where $D \gg 10^3$, evaluating these dense candidate masks forces wrapper classifiers to compute pairwise distances across thousands of non-informative dimensions at a computational complexity scaling as $O(N_{\text{samp}}^2 d)$, which squanders early function evaluation budgets and precipitates severe classifier overfitting under limited sample observations, particularly in sample-scarce biomedical microarrays where observations $N_{\text{samp}} \ll 100$. To establish an informative yet diverse initial swarm while systematically mitigating mid-evolution search stagnation, KDS-EL introduces a prior-guided sparse population initialization strategy coupled with an adaptive stagnation recovery mechanism, as systematically illustrated across the four operational stages in Fig. 3.

3.2.1. Prior-guided sparse population initialization

Given the training dataset instance $\mathcal{D} = \{(\mathbf{f}_k, y_k)\}_{k=1}^{N_{\text{samp}}}$ with continuous feature vectors $\mathbf{f}_k = [f_{k,1}, \ldots, f_{k,D}] \in \mathbb{R}^D$ and discrete class labels $y_k \in \{1, \ldots, C\}$, the non-linear dependencies between individual candidate dimensions and target categories are quantified through continuous-discrete mutual information [39]. To handle continuous real-valued inputs without relying on arbitrary data discretization, the mutual information score $\hat{I}(f_j; Y)$ for each feature $j \in \{1, \ldots, D\}$ is computed via the continuous-discrete k-nearest neighbors KSG estimator formulated by Ross [39] as an adaptation of Kraskov et al. [38]:

$$\hat{I}(f_j; Y) = \psi(N_{\text{samp}}) - \frac{1}{N_{\text{samp}}} \sum_{k=1}^{N_{\text{samp}}} \psi(N_{y_k}) + \psi(k_{\text{nn}}) - \frac{1}{N_{\text{samp}}} \sum_{k=1}^{N_{\text{samp}}} \psi(n_{f_j,k}), \quad (12)$$

where $\psi(\cdot)$ denotes the digamma function, N_{y_k} is the count of training samples sharing class label y_k, $k_{\text{nn}} = 3$ is the neighborhood parameter in the joint continuous-discrete space, and $n_{f_j,k}$ is the count of data points whose marginal continuous distance to sample k along feature f_j is strictly smaller than the distance to its k_{nn}-th neighbor in the class-conditioned subspace. Because mutual information is theoretically non-negative, finite-sample stochastic fluctuations yielding negative numerical values are truncated via $\max(0, \hat{I}(f_j; Y))$. To establish a standardized probability profile across heterogeneous datasets,

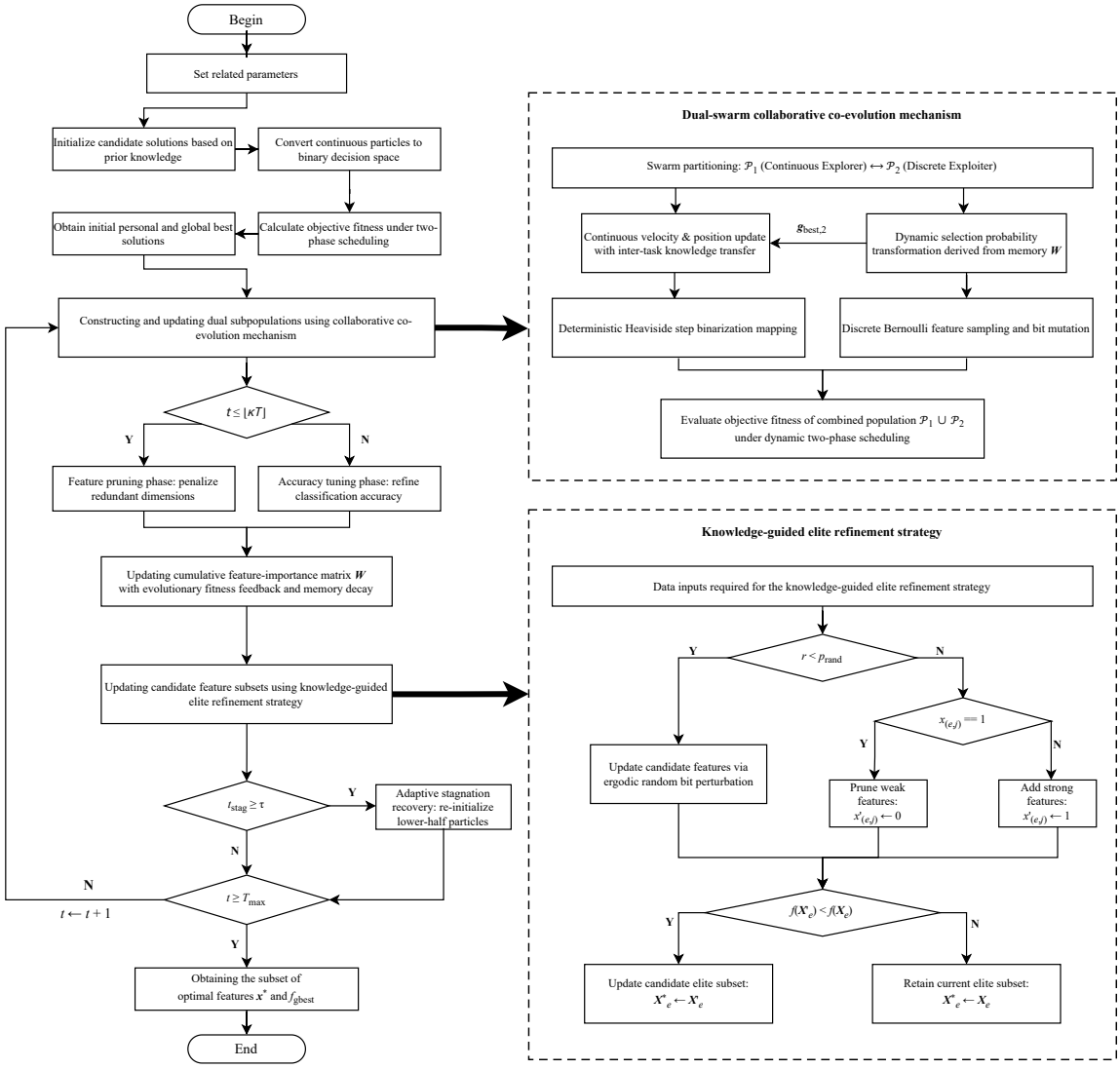


Figure 2: Execution flowchart of the collaborative dual-swarm KDS-EL framework.

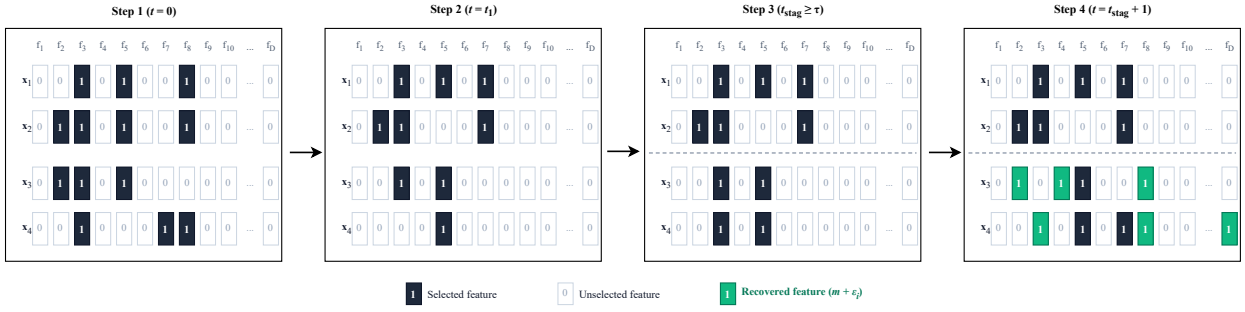


Figure 3: Schematic of prior-guided sparse population initialization and adaptive diversity recovery.

these raw mutual information scores are normalized into a prior relevance vector $\mathbf{m} = [m_1, \ldots, m_D] \in [0,1]^D$:

$$m_j = \frac{\hat{I}(f_j; Y) - \min_{1 \le l \le D} \hat{I}(f_l; Y)}{\max_{1 \le l \le D} \hat{I}(f_l; Y) - \min_{1 \le l \le D} \hat{I}(f_l; Y) + \epsilon}, \quad j = 1, \ldots, D, \tag{13}$$

where $\epsilon > 0$ is a small numerical constant preventing division by zero.

In algorithmic formulation, the mutual information prior calculation operates strictly on the data partition presented to the optimizer. Following the established benchmark instance protocol of peer evolutionary feature selection literature [13, 9, 12], each evaluated dataset $\mathcal{D}$ serves as a self-contained combinatorial optimization instance where the prior relevance vector $\mathbf{m}$ is computed once offline across the instance to provide heuristic domain guidance, while an internal 5-fold stratified cross-validation scheme serves as the wrapper fitness evaluation surrogate. In prospective inductive clinical deployments with external independent cohorts, this prior computation is strictly encapsulated within the training partition to guarantee unbiased out-of-fold generalization.

To prevent population homogenization while guiding search trajectories toward informative dimensions, candidate particles are initialized through a hybrid seeding scheme parameterized by a seeding ratio $\rho \in (0,1)$, as illustrated

in Step 1 of Fig. 3. Continuous position vectors $\mathbf{p}_i = [p_{i,1}, \ldots, p_{i,D}] \in [0,1]^D$ are constructed as:

$$\mathbf{p}_i = \begin{cases} \text{clip}\left(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1\right), & 1 \le i \le \lfloor \rho N \rfloor, \\ \mathbf{u}_i, & \lfloor \rho N \rfloor < i \le N, \end{cases} \tag{14}$$

where $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ represents zero-mean Gaussian perturbation with standard deviation $\sigma = 0.10$, $\mathbf{u}_i \sim \mathcal{U}(0,1)^D$ provides uniform stochastic coverage across unguided dimensions, and $\text{clip}(z, 0, 1) = \max(0, \min(1, z))$ restricts continuous coordinates to the unit hypercube.

Continuous position vectors $\mathbf{p}_i$ are subsequently transformed into binary decision vectors $\mathbf{x}_i = [x_{i,1}, \ldots, x_{i,D}] \in \{0,1\}^D$ via a deterministic Heaviside step mapping with threshold $\theta \in (0.5, 1)$:

$$x_{i,j} = \begin{cases} 1, & \text{if } p_{i,j} \ge \theta, \\ 0, & \text{otherwise}, \end{cases} \quad j = 1, \ldots, D. \tag{15}$$

By setting the binarization threshold to $\theta = 0.60 \in (0.5, 1)$, candidate features are activated with an expected probability of 0.40 for unguided particles, breaking the symmetric 0.50 expectation. For prior-seeded particles, this threshold restricts initial activation strictly to candidate dimensions with a normalized relevance score exceeding 0.60, producing a sparse initial population that mitigates initial search blindness and accelerates early convergence toward discriminative subspaces.

Estimation robustness and mitigation of the Winner's curse under sample scarcity: In ultra-high-dimensional and sample-scarce regimes where candidate dimensions $D \gg 10^3$ vastly exceed sample observations $N_{\text{samp}} \ll 100$ by two to three orders of magnitude, non-parametric entropy estimation on limited observations inherently exhibits non-trivial finite-sample variance. Under extensive dimensional testing, stochastic sample alignment across thousands of candidate features may cause irrelevant noise dimensions to obtain spuriously elevated mutual information estimates, reflecting a classic manifestation of the Winner's curse in high-dimensional screening. If subjected to standard Min-Max normalization, such estimation fluctuations risk being amplified toward $m_j \approx 1$, potentially injecting false-positive biases into the initial population.

Crucially, KDS-EL establishes a multi-layered architectural defense that insulates the evolutionary search against noisy prior contamination. First, KDS-EL fundamentally departs from conventional filter pre-selectors by rejecting irreversible hard feature truncation; rather than permanently discarding unselected dimensions, the normalized vector $\mathbf{m}$ establishes a continuous soft prior manifold, maintaining the complete $\mathbb{R}^D$ feature space fully accessible throughout optimization. Second, the hybrid initialization in (14) injects Gaussian dispersion $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ around $\mathbf{m}$ while maintaining an unguided sub-swarm $\mathbf{u}_i \sim \mathcal{U}(0,1)^D$ across $(1-\rho)N$ particles, preventing premature population collapse into false-positive empirical peaks. Third, every candidate subset is immediately subjected to wrapper cross-validation classification feedback; because spurious noise features fail to sustain cross-validation classification consistency, they generate zero or negative fitness differentials. Under the exponential memory decay governed by decay factor γ in (20), their transient weights in the cumulative feature-importance vector $\mathbf{W} \in \mathbb{R}^D$ rapidly decay to zero. Consequently, the knowledge-guided elite refinement mechanism swiftly purges these non-contributing dimensions in favor of synergistic, discriminative features. Finally, should the swarm become temporarily trapped in empirical local attractors, the adaptive stagnation recovery routine (Section 3.2.2) re-samples the stagnant subpopulation under fresh stochastic perturbations, ensuring robust long-term exploratory self-correction.

3.2.2. Adaptive stagnation recovery

In iterative swarm search over ultra-high-dimensional fitness landscapes, population diversity inevitably decays as particles converge toward shared historical attractors. As evolutionary generations progress, particle velocities decelerate toward zero, causing the swarm to cluster tightly within confined local basins. In high-dimensional regimes where candidate dimensions $D \gg 10^3$ vastly exceed available training observations N_{samp}, such velocity decay precipitates premature search stagnation, trapping the optimizer in sub-optimal feature combinations governed by empirical noise. Conventional meta-heuristics frequently attempt to combat stagnation via uniform random re-initialization or heavy-tailed mutation perturbations, such as Cauchy mutations or opposition-based learning [15, 16]. However, in ultra-high-dimensional spaces, uniform random restarts inevitably collapse candidate solutions back into dense, uninformative subspaces, squandering subsequent wrapper evaluations on non-informative noise dimensions. Similarly, blind heavy-tailed mutations in 2^D combinatorial space perturb candidate bit positions without domain guidance, dismantling carefully discovered, highly cooperative feature combinations with an overwhelming probability exceeding 99.9%. Furthermore, classical opposition-based learning strategies invert binary masks across all dimensions, transforming sparse subsets into massive complementary masks of cardinality $D - d \approx 10^4$, thereby exacerbating classifier overfitting and computational latency. Consequently, sustaining exploratory diversity in ultra-high dimensions requires an asymmetric recovery mechanism capable of injecting targeted stochastic exploration while preserving verified elite solutions.

To resolve this structural dilemma without incurring catastrophic forgetting or dimensional bloat, KDS-EL introduces an adaptive stagnation recovery mechanism. The adaptiveness of this mechanism stems from its feedback-driven, on-demand activation: unlike classical heuristic restarts that disrupt search trajectories on fixed generational schedules regardless of convergence status, KDS-EL couples recovery execution dynamically to runtime fitness stagnation feedback. Throughout generational search, a stagnation counter t_{stag} monitors the consecutive generational span during which the global-best fitness fails to achieve a strict numerical decrease. As long as the swarm sustains productive descent, the counter t_{stag} is continuously reset, leaving ongoing trajectories undisturbed; recovery is triggered strictly on demand when $t_{\text{stag}} \ge \tau$, where $\tau \in \mathbb{N}^+$ serves as a preset patience parameter regulating detection sensitivity. When this condition is met, search stagnation is formally identified, triggering an asymmetric bipartite population bisection, as illustrated in Step 3 of Fig. 3. Specifically, candidate particles are ranked in ascending order according to their historical personal-best fitness values $f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t))$. To safeguard converged, highly discriminative feature combinations, the superior half of the swarm comprising $\lfloor N/2 \rfloor$ particles is preserved intact, maintaining the optimization memory accumulated along successful search trajectories. Concurrently, the lower-performing half comprising the remaining $N - \lfloor N/2 \rfloor$ stagnant particles has its continuous positions $\mathbf{p}_i$ refreshed by re-sampling around the prior relevance vector $\mathbf{m} \in [0,1]^D$ under Gaussian perturbation, with its binary personal-best memory vector $\mathbf{p}_{\text{best},i} \in \{0,1\}^D$ synchronized to the resulting candidate decision mask $\mathbf{x}_i$:

$$\begin{aligned} &\mathbf{p}_i \leftarrow \text{clip}\left(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1\right), \\ &\mathbf{x}_i \leftarrow \mathbb{I}(\mathbf{p}_i \ge \theta), \quad \mathbf{p}_{\text{best},i} \leftarrow \mathbf{x}_i, \end{aligned} \tag{16}$$

where $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ represents zero-mean Gaussian perturbation with standard deviation $\sigma = 0.10$, $\mathbb{I}(\cdot)$ denotes element-wise Heaviside thresholding, $\theta \in (0.5, 1)$ denotes the cut-off binarization threshold, and $\text{clip}(z, 0, 1) = \max(0, \min(1, z))$ restricts continuous coordinates to the unit hypercube.

Immediately following position refreshing, the candidate binary mask $\mathbf{x}_i$ is evaluated via the internal cross-validation wrapper under current objective weights $\mathbf{w}(t)$ before updating the personal-best record $f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t)) \leftarrow f(\mathbf{x}_i, \mathbf{w}(t))$, as depicted in Step 4 of Fig. 3. The accepted mask is subsequently submitted to generational evolution, ensuring that coordinate differentials $C_i^{(t)}$ in importance learning are calculated strictly against the established historical personal best, while the stagnation counter is reset to zero ($t_{\text{stag}} \leftarrow 0$). Crucially, this bipartite design bounds the additional computational overhead to at most $\lfloor N/2 \rfloor$ wrapper evaluations, which are invoked strictly on demand rather than continuously incurred across every generation. Furthermore, because re-sampling is anchored to the continuous prior relevance vector $\mathbf{m}$ rather than uniform noise, newly generated candidate masks remain structurally sparse, entirely circumventing unguided dimensional expansion. When coupled with the exponential memory forgetting of non-contributing dimensions in the cumulative feature-importance vector $\mathbf{W} \in \mathbb{R}^D$, this adaptive recovery mechanism breaks exploratory deadlocks, injects fresh variance into dormant subspaces, and propels the swarm toward globally competitive, sparse feature subsets.

3.3. Cumulative importance learning and elite refinement

To overcome the search amnesia and destructive perturbation bottlenecks of conventional meta-heuristics, KDS-EL establishes a closed-loop knowledge learning mechanism comprising online importance accumulation and targeted directional elite refinement.

In ultra-high-dimensional feature selection where candidate dimensions $D \gg 10^3$ vastly exceed sample observations, traditional evolutionary wrappers rely on undirected stochastic bit flips with a uniform mutation probability $p_{\text{mut}} = 1/D$. Under this blind perturbation regime, meta-heuristics inevitably suffer from two structural search lesions. First, when an established, highly informative feature with active status $x_{e,j} = 1$ is blindly mutated to 0, delicate multi-feature synergies are dismantled, inducing severe classification degradation. Second, when an uninformative background noise feature with inactive state $x_{e,j} = 0$ is randomly flipped to 1, the candidate subset expands unnecessarily, escalating evaluation latency and precipitating classifier overfitting. Be-

cause the probability of randomly hitting informative dimensions within a 2^D combinatorial neighborhood is asymptotically infinitesimal, scaling as $O(1/D)$, undirected mutation squanders function evaluations on destructive trials. Furthermore, conventional meta-heuristics exhibit severe search amnesia, as historical fitness feedback from intermediate iterations is discarded immediately after particle position updates.

3.3.1. Online feature importance learning

In ultra-high-dimensional wrapper feature selection, standard evolutionary algorithms suffer from severe search amnesia. Although thousands of candidate feature subsets are iteratively synthesized and evaluated via cross-validation, historical fitness differentials produced by coordinate transitions are discarded immediately following generational updates. Consequently, evolutionary trajectories remain memoryless, repeatedly re-evaluating uninformative dimensions and barren combinatorial regions. To bridge generational evaluations with cumulative historical learning, KDS-EL maintains a persistent cumulative directional credit vector $\mathbf{W} = [W_1, \ldots, W_D] \in \mathbb{R}^D$, initialized to $\mathbf{0}$, which systematically records the directional contribution of each candidate dimension toward classification discriminability.

At generation t, whenever an updated candidate solution $\mathbf{x}_i^{(t)} \in \{0,1\}^D$ achieves a strict fitness improvement over its historical personal best $\mathbf{p}_{\text{best},i} \in \{0,1\}^D$, the algorithm isolates the differential feature set $C_i^{(t)} \subset \{1, \ldots, D\}$ containing coordinates whose binary states were altered during the evolutionary update:

$$C_i^{(t)} = \left\{ j \in \{1, \ldots, D\} \;\middle|\; x_{i,j}^{(t)} \neq p_{\text{best},i,j} \right\}. \tag{17}$$

Let $\delta_i^{(t)} = f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t)) - f(\mathbf{x}_i^{(t)}, \mathbf{w}(t))$ denote the generational fitness differential evaluated under dynamic weight vector $\mathbf{w}(t)$. To distinguish beneficial feature alterations from detrimental ones, the algorithm executes bidirectional credit updating. When candidate solution $\mathbf{x}_i^{(t)}$ achieves a strict fitness improvement over its historical personal best with $\delta_i^{(t)} > 0$, each altered coordinate $j \in C_i^{(t)}$ receives a directional reinforcement scaled by $\delta_i^{(t)}$:

$$W_j \leftarrow W_j + \delta_i^{(t)} \cdot (2x_{i,j}^{(t)} - 1), \quad \forall j \in C_i^{(t)}, \tag{18}$$

where the polarity factor $(2x_{i,j}^{(t)} - 1) \in \{-1, +1\}$ yields $+1$ when feature j was newly activated with $x_{i,j}^{(t)} = 1$ to achieve the superior solution, reinforcing its importance score, and yields -1 when feature j was successfully pruned with $x_{i,j}^{(t)} = 0$, penalizing its inclusion. Conversely, when candidate solution $\mathbf{x}_i^{(t)}$ fails to improve upon the personal best, characterized by $\Delta_i^{(t)} = f(\mathbf{x}_i^{(t)}, \mathbf{w}(t)) - f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t)) \geq 0$, the coordinates altered during the unrewarded transition incur a symmetric penalty:

$$W_j \leftarrow W_j - \Delta_i^{(t)} \cdot (2x_{i,j}^{(t)} - 1), \quad \forall j \in C_i^{(t)}. \tag{19}$$

This bidirectional credit-penalty dynamic ensures that beneficial transitions reinforce feature utility while deteriorating moves dynamically suppress counterproductive coordinate changes.

Because candidate subsets become progressively sparser across evolution, unconstrained accumulation would allow early-discovered, coarsely discriminative dimensions to dominate $\mathbf{W}$ indefinitely. Such historical inertia suppresses emergent synergistic feature interactions that only become apparent once redundant dimensions are pruned. To reconcile long-term knowledge retention with responsiveness to emergent feature synergies, an exponential memory decay is applied at the conclusion of each generation:

$$\mathbf{W} \leftarrow \gamma \cdot \mathbf{W}, \tag{20}$$

where $\gamma \in (0,1)$ denotes the continuous memory retention factor governing the exponential forgetting rate. Consequently, dimensions that consistently yield fitness gains sustain prominent positive scores, whereas transient or uninformative coordinates undergo geometric attenuation toward zero. Crucially, as $\gamma \to 1$, historical search credit accumulates without forgetting, risking exploratory fixation; conversely, as $\gamma \to 0$, memory collapses into single-step myopic feedback. Maintaining $\gamma \in (0,1)$ ensures that the importance vector provides a calibrated probabilistic landscape that directly steers discrete sampling in Subpopulation $\mathcal{P}_2$ and guides the directional elite refinement strategy.

3.3.2. Knowledge-guided elite refinement

In ultra-high-dimensional regimes where candidate dimensions $D \gg 10^3$ vastly exceed sample observations, uniform bit-flipping degenerates into blind trial-and-error, disrupting synergistic feature coalitions and precipitating search stagnation. Moreover, applying local search across all particles incurs prohibitive computational latency scaling with swarm size N. To overcome this combinatorial dilemma, the KER strategy exploits the continuous feature-importance landscape $\mathbf{W}$ to steer directional bit inversions strictly on a compact elite echelon $k_e \ll N$, concentrating search momentum on high-potential dimensions while bounding evaluation overhead to exactly k_e fitness calculations per generation.

Algorithm 1 Knowledge-guided elite refinement strategy

Require: Elite candidate solution $\mathbf{x}_e$, importance vector $\mathbf{W}$, budget k_{flip}, exploration probability p_{rand}, objective weights $\mathbf{w}(t)$
Ensure: Refined candidate solution $\mathbf{x}_e^\star$
Branch 1: Ergodic random exploration (with probability p_{rand})
  Uniformly sample a coordinate subset $\mathcal{S}_{\text{rand}}$ of size k_{flip};
  Invert binary coordinates in $\mathbf{x}_e$ over $\mathcal{S}_{\text{rand}}$ via Eq. (21);
Branch 2: Knowledge-guided directional refinement (with probability $1 - p_{\text{rand}}$)
  Calculate directional perturbation scores $\boldsymbol{\sigma}$ using Eq. (23);
  Select top-k_{flip} candidate feature subset $\mathcal{S}_{\text{know}}$ by Eq. (24);
  Invert binary coordinates in $\mathbf{x}_e$ over $\mathcal{S}_{\text{know}}$;
Evaluate mutant candidate fitness using wrapper classifier by Eq. (27);
Perform greedy pairwise in-place replacement according to Eq. (25);
return Refined elite candidate solution $\mathbf{x}_e^\star$;

At each generation, the top k_e individuals ranked by objective fitness are admitted into the elite pool. For each elite candidate $\mathbf{x}_e \in \{0,1\}^D$, the number of perturbed features is set to $k_{\text{flip}} = \max(1, \lfloor \phi D \rfloor)$, where $\phi \in (0,1)$ represents the perturbation fraction. As depicted in the operational schematic of Fig. 4, the KER strategy probabilistically executes one of two operational pathways:

Branch 1: Ergodic random exploration: To preserve search ergodicity and prevent the elite pool from stagnating in deceptive local basins, a feature index subset $\mathcal{S}_{\text{rand}} \subset \{1, \ldots, D\}$ of cardinality k_{flip} is sampled uniformly at random with exploration probability $p_{\text{rand}} \in [0,1]$, and the corresponding coordinates are inverted:

$$x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{rand}}, \quad |\mathcal{S}_{\text{rand}}| = k_{\text{flip}}. \tag{21}$$

Branch 2: Knowledge-guided directional refinement: With complementary probability $1 - p_{\text{rand}}$, the strategy executes targeted directional refinement driven by the cumulative importance landscape. For each candidate dimension $j \in \{1, \ldots, D\}$, a state-dependent directional flip score $\sigma_j \in \mathbb{R}$ is formulated as:

$$\sigma_j = (1 - 2x_{e,j}) \cdot W_j = \begin{cases} +W_j, & \text{if } x_{e,j} = 0, \\ -W_j, & \text{if } x_{e,j} = 1. \end{cases} \tag{22}$$

A positive flip score $\sigma_j > 0$ reflects two favorable search transitions: an unselected feature with inactive coordinate $x_{e,j} = 0$ that exhibits a positive historical contribution $W_j > 0$, or an active feature with $x_{e,j} = 1$ whose exclusion has historically improved classification fitness with $W_j < 0$. In either scenario, flipping the binary state of feature j is expected to enhance candidate solution quality.

To balance directional guidance with exploratory diversity and resolve potential ties among identical scores, an adaptive stochastic perturbation $\xi_j \sim \mathcal{N}\left(0, (\text{std}(\mathbf{W}) + 10^{-5})^2\right)$ proportional to the empirical dispersion of the learned importance vector is injected into each directional score:

$$\sigma_j = (1 - 2x_{e,j}) \cdot W_j + \xi_j, \quad \xi_j \sim \mathcal{N}\left(0, \left(\text{std}(\mathbf{W}) + 10^{-5}\right)^2\right). \tag{23}$$

The feature subset $\mathcal{S}_{\text{know}}$ containing the k_{flip} coordinates with the largest perturbed directional scores is identified:

$$\mathcal{S}_{\text{know}} = \arg \max_{\substack{\mathcal{S} \subseteq \{1, \ldots, D\} \\ |\mathcal{S}| = k_{\text{flip}}}} \sum_{j \in \mathcal{S}} \sigma_j, \tag{24}$$

and the corresponding binary states are inverted via $x'_{e,j} \leftarrow 1 - x_{e,j}$ for all $j \in \mathcal{S}_{\text{know}}$.

Greedy pairwise in-place replacement: The perturbed candidate $\mathbf{x}'_e$ is evaluated using the objective fitness function $f(\mathbf{x}'_e, \mathbf{w}(t))$. A greedy pairwise in-place replacement rule determines solution admission:

$$\mathbf{x}_e^\star = \begin{cases} \mathbf{x}'_e, & \text{if } f(\mathbf{x}'_e, \mathbf{w}(t)) < f(\mathbf{x}_e, \mathbf{w}(t)), \\ \mathbf{x}_e, & \text{otherwise.} \end{cases} \tag{25}$$

Upon achieving a strict fitness improvement, $\mathbf{x}'_e$ immediately replaces $\mathbf{x}_e$ in the population and updates the global-best solution if $f(\mathbf{x}'_e, \mathbf{w}(t)) < f_{\text{gbest}}$. Because the elite pool cardinality $k_e \ll N$ comprises only a minor fraction of the overall swarm, the KER strategy operates within a controlled computational overhead

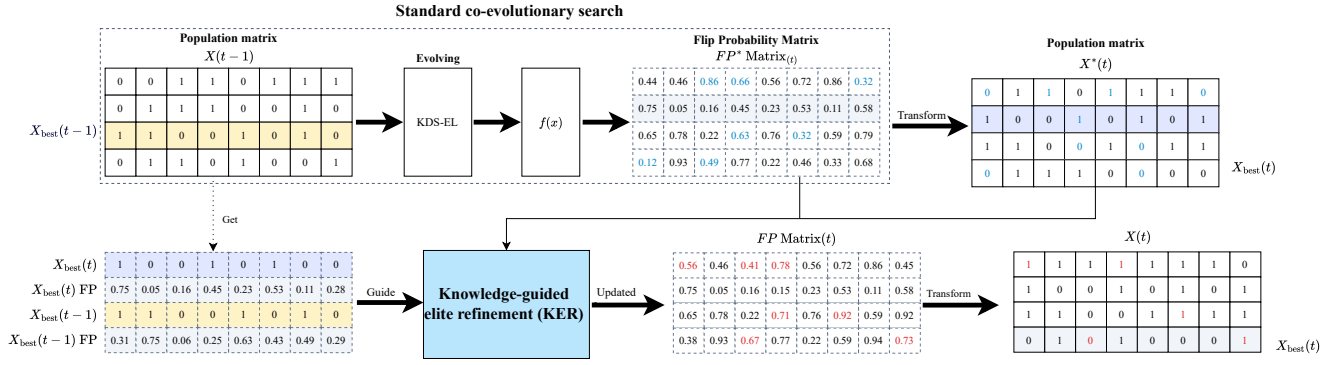


Figure 4: Operational mechanism of the knowledge-guided elite refinement strategy.

of k_e fitness evaluations per generation. The complete procedural logic of elite refinement is formalized in Algorithm 1.

Dual-scale search dynamics and dimension-scaling analysis: The perturbation budget $k_{\text{flip}} = \max(1, \lfloor \phi D \rfloor)$ scales with ambient dimensionality D, producing distinct search behaviors across problem scales. In moderate-dimensional regimes where $D \leq 2{,}000$, k_{flip} produces small-scale bit flips that execute fine-grained local neighborhood refinement around elite solutions. Conversely, in ultra-high-dimensional regimes where ambient dimensionality expands past $D > 10^4$, the perturbation budget scales proportionally into hundreds of simultaneous coordinate transitions, causing the KER strategy to shift dynamically from localized perturbation into a guided macro-recombination mechanism. Operating at this broader structural scale allows the swarm to evaluate coordinated multi-feature activations and deactivations guided by $\mathbf{W}$, facilitating transitions across distant combinatorial basins. Crucially, the greedy pairwise in-place replacement rule in (25) serves as an intrinsic safeguard against destructive mutations: whenever an extensive macro-flip degrades classification accuracy or inflates feature penalties, the candidate mutant is discarded without modifying the elite pool. While fixed-fraction scaling $\lfloor \phi D \rfloor$ performs consistently across diverse benchmark scales, adaptively modulating the perturbation budget according to the active subset cardinality $d_{\text{selected}} = \|\mathbf{x}_e\|_0$ via $k_{\text{flip}} = \max(1, \lfloor \phi \cdot \|\mathbf{x}_e\|_0 \rfloor)$, or enforcing an upper saturation cap $k_{\max}$, represents an effective adaptive refinement for extreme-dimensional applications.

3.4. Dynamic two-phase weight scheduling

In ultra-high-dimensional wrapper feature selection where candidate dimensions $D \gg 10^3$ vastly exceed sample observations N_{samp}, optimization must balance classification error against subset cardinality. Conventional metaheuristics rely on static scalarization, inducing severe search degeneration. A conservative feature penalty fails to overcome the dense dimensionality attractor, squandering evaluation budgets and precipitating classifier overfitting. Conversely, maintaining an elevated penalty throughout evolution suppresses late-stage convergence, prematurely discarding synergistic feature coalitions whose retention slightly increases cardinality but is vital for resolving complex decision boundaries.

To resolve this temporal conflict, KDS-EL introduces the dynamic TPS strategy, decoupling the search horizon into two functional phases, as depicted in the operational flowchart of Fig. 2. The total generation budget T is partitioned at a split point parameterized by the phase transition ratio $\kappa \in (0, 1)$. The dynamic objective weight vector $\mathbf{w}(t) = [w_a(t), w_f(t)] \in [0, 1]^2$, satisfying the normalization condition $w_a(t) + w_f(t) = 1$, is formulated as:

$$\mathbf{w}(t) = \begin{cases} [w_{a,1}, w_{f,1}], & 1 \leq t \leq \lfloor \kappa T \rfloor, \\ [w_{a,2}, w_{f,2}], & \lfloor \kappa T \rfloor < t \leq T, \end{cases} \tag{26}$$

where the weight coefficients satisfy the strict ordering $w_{f,1} > w_{f,2}$ and $w_{a,1} < w_{a,2}$. At generation t, the unified scalar fitness $f(\mathbf{x}, \mathbf{w}(t))$ of a candidate binary mask $\mathbf{x} \in \{0, 1\}^D$ is evaluated under an exterior penalty barrier that safeguards classifier integrity:

$$f(\mathbf{x}, \mathbf{w}(t)) = \begin{cases} 1.0 + 0.1 \cdot (d_{\min} - \|\mathbf{x}\|_0), & \text{if } \|\mathbf{x}\|_0 < d_{\min}, \\ w_a(t) \cdot (1 - \text{Acc}(\mathbf{x})) + w_f(t) \cdot \left(\dfrac{\|\mathbf{x}\|_0}{D} \right), & \text{if } \|\mathbf{x}\|_0 \geq d_{\min}, \end{cases} \tag{27}$$

where $\text{Acc}(\mathbf{x}) \in [0, 1]$ denotes the 5-fold cross-validation accuracy evaluated by a k-NN classifier with neighborhood parameter $k_{\text{nn}} = 3$, $\|\mathbf{x}\|_0 = \sum_{j=1}^{D} x_j$ quantifies active feature cardinality, and $d_{\min} = 3$ denotes the minimum subset cardinality constraint. To prevent evolutionary search from collapsing into trivial or uninformative low-dimensional projections with insufficient multivariate discriminative capacity, the exterior penalty assigns a fitness value strictly exceeding 1.0 whenever $\|\mathbf{x}\|_0 < d_{\min}$, exerting strong selection pressure that promptly eliminates sub-minimal configurations while safeguarding search continuity.

The two evolutionary regimes execute complementary optimization roles. Throughout Phase 1, spanning $1 \leq t \leq \lfloor \kappa T \rfloor$, the elevated feature penalty $w_{f,1} = 0.30$ with $w_{a,1} = 0.70$ exerts substantial pruning pressure against non-informative features, rapidly contracting candidate dimensionality to a compact fraction of D. In Phase 2, spanning $\lfloor \kappa T \rfloor < t \leq T$, weights shift to $w_{a,2} = 0.95$ and $w_{f,2} = 0.05$ under split ratio $\kappa = 0.70$, reorienting evolutionary pressure toward fine classification refinement. Because objective weights transition abruptly at generation $t = \lfloor \kappa T \rfloor + 1$, candidate fitness values operate on disparate scales. To mitigate scale distortion and preserve search continuity, all personal-best solutions are instantaneously re-evaluated under Phase 2 weights, setting $f_{\text{pbest},i} \leftarrow f(\mathbf{p}_{\text{best},i}, \mathbf{w}_2)$ for all $i \in \{1, \ldots, N\}$, updating $\mathbf{g}_{\text{best}}$, and resetting stagnation counter $t_{\text{stag}} \leftarrow 0$. This boundary calibration safeguards memory integrity, ensuring Phase 2 builds cohesively upon Phase 1 pruning without numerical distortion.

3.5. Overall algorithmic procedure and complexity analysis

Algorithm 2 synthesizes the overall execution pipeline of KDS-EL into a unified multi-stage optimization framework. The complete workflow cohesively coordinates offline filter-level relevance extraction, knowledge-informed hybrid seeding, dynamic two-phase objective scheduling with boundary memory calibration, collaborative dual-subpopulation co-evolution, differential importance harvesting, targeted elite refinement, and adaptive stagnation recovery.

Algorithm 2 Overall framework of KDS-EL

Require: Training dataset $\mathcal{D}$, population size N, maximum generations T, parameters $(\theta, \rho, \sigma, \gamma, k_e, \phi, p_{\text{rand}}, \kappa, \tau)$
Ensure: Optimal compact feature subset $\mathbf{x}^\star$ and optimal fitness f_{gbest}
Stage 1: Filter-level relevance extraction
Extract and normalize mutual information prior vector $\mathbf{m}$ using Eqs. (12) and (13);
Stage 2: Hybrid population seeding
Generate initial swarm $\mathcal{P}$ by combining MI-guided Gaussian seeds with uniform particles via Eqs. (14)–(15);
Initialize importance vector $\mathbf{W} \leftarrow \mathbf{0}$, initial weights $\mathbf{w}(1)$, and evaluate candidate fitness by Eq. (27);
Stage 3: Dynamic two-phase co-evolutionary loop
for $t = 1$ **to** T **do**
  Schedule dynamic objective weights $\mathbf{w}(t)$ according to Eq. (26);
  if $t = \lfloor \kappa T \rfloor + 1$ **then** calibrate personal and global bests under Phase 2 weights;
  Update $\mathcal{P}_1$ continuous positions by Eq. (10) and sample $\mathcal{P}_2$ discrete masks by Eq. (11);
  Evaluate candidate fitness $f(\mathbf{x}_i^{(t)}, \mathbf{w}(t))$ by Eq. (27) under exterior cardinality penalty;
  Update personal bests and credit/penalize $\mathbf{W}$ over $C_i^{(t)}$ via Eqs. (18)–(20);
  Refine top k_e elite candidates using the KER strategy according to Algorithm 1;
  Update global best $\mathbf{g}_{\text{best}}$ and monitor swarm stagnation counter;
  if stagnation persists for τ generations **then** execute adaptive recovery by Eq. (16);
end for
return Optimal feature subset $\mathbf{x}^\star \leftarrow \mathbf{g}_{\text{best}}$ and optimal fitness f_{gbest};

3.5.1. Computational complexity analysis

The computational overhead of KDS-EL decouples into one-time offline preprocessing and online generational evolution. Offline mutual information estimation and prior relevance ranking across D dimensions on N_{samp} training samples consume $\mathcal{T}_{\text{pre}} = O(D \cdot N_{\text{samp}} \log N_{\text{samp}} + D \log D)$ operations. During online evolution across T generations, computational expenditure encompasses four constituent components: (i) dual-subpopulation position-velocity updates requiring $O(N \cdot D)$ operations per generation; (ii) wrapper fitness evaluations across N candidate solutions consuming $O(N \cdot N_{\text{samp}}^2 \cdot \bar{d}^{(t)})$ Euclidean distance calculations at generation t, where $\bar{d}^{(t)} = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{x}_i\|_0$; (iii) cumulative importance learning and KER refinement requiring $O(N \cdot D)$ memory updating, $O(D \log D)$ directional score ranking, and $O(k_e \cdot N_{\text{samp}}^2 \cdot \bar{d}_{\text{elite}}^{(t)})$ elite re-evaluations; and (iv) adaptive stagnation recovery re-evaluating $\lfloor N/2 \rfloor$ particles when triggered. Summing all operational components across T generations yields the complete unsimplified asymptotic time upper bound of KDS-EL:

$$\begin{aligned}\mathcal{T}_{\text{comp}} = O\Big(&\mathcal{T}_{\text{pre}} + T(N + k_e)N_{\text{samp}}^2 \bar{d}_{\text{selected}} \\ &+ T(N \cdot D + D \log D) + N_{\text{rec}} \lfloor N/2 \rfloor N_{\text{samp}}^2 \bar{d}_{\text{rec}}\Big),\end{aligned} \tag{28}$$

where $\bar{d}_{\text{selected}}$ denotes the average active feature count across generations, and $N_{\text{rec}} \leq \lfloor T/\tau \rfloor$ denotes the maximum stagnation recovery invocations. Because the elite pool size is bounded by a fixed small constant $k_e = 3 \ll N$, we have $(N + k_e) = O(N)$. Furthermore, given that stagnation recovery is invoked sparsely ($N_{\text{rec}} \leq 5 \ll T$ under $\tau = 20$), its computational contribution is asymptotically subsumed by generational evaluations. Consequently, the dominant asymptotic time complexity of KDS-EL simplifies to:

$$\mathcal{T}_{\text{total}} = O\left(T \cdot N \cdot N_{\text{samp}}^2 \cdot \bar{d}_{\text{selected}} + T(N \cdot D + D \log D)\right). \tag{29}$$

Although KDS-EL shares the identical worst-case asymptotic upper bound $O(T \cdot N \cdot N_{\text{samp}}^2 \cdot D)$ with canonical dual-swarm frameworks when expressed in terms of ambient dimensionality D, KDS-EL achieves an empirical 3.4-fold wall-clock acceleration, averaging 51.3 s compared to 175.0 s across the benchmark suite. This pronounced divergence between asymptotic complexity order and empirical execution latency stems directly from the cooperative interplay between prior-guided sparse initialization and dynamic Two-Phase Weight Scheduling. Rather than evaluating dense candidate masks where unguided swarms retain an average of over 2,700 features, KDS-EL aggressively contracts active feature cardinality to an ultra-sparse representation averaging merely 87.8 dimensions from early generations. Because computing pairwise Euclidean distances in the k-NN classifier scales strictly linearly with active feature count d_{selected}, contracting selected dimensions by orders of magnitude directly slashes the actual floating-point distance calculations per wrapper evaluation, converting theoretical subset sparsity into pronounced real-world wall-clock acceleration.

Regarding spatial footprint, memory consumption comprises the binary population $\mathbf{X} \in \{0,1\}^{N \times D}$, continuous positions $\mathbf{P} \in [0,1]^{N \times D}$, velocities $\mathbf{V} \in \mathbb{R}^{\lfloor N/2 \rfloor \times D}$, personal bests $\mathbf{P}_{\text{best}} \in \{0,1\}^{N \times D}$, feature importance memory $\mathbf{W} \in \mathbb{R}^D$, and training data $\mathcal{D} \in \mathbb{R}^{N_{\text{samp}} \times D}$. The resulting total spatial complexity is $\mathcal{S}_{\text{total}} = O\big((3N + N_{\text{samp}} + 1)D\big) = O\big((N + N_{\text{samp}})D\big)$, scaling strictly linearly with feature dimensionality D. Because allocations utilize standard contiguous numerical arrays with small constant factors, working memory scales strictly linearly and remains modest even for high-dimensional problem instances with $D > 10^5$, ensuring computationally feasible execution within standard workstation memory budgets.

4. Experimental results and discussion

This section outlines the experimental framework and presents a comprehensive empirical evaluation of KDS-EL across eight analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects. All aggregate numbers are computed over the 20 benchmark datasets described in Section 4.1.

4.1. Experimental setup

This subsection outlines the experimental framework designed to evaluate the classification performance, dimensionality reduction capability, computational efficiency, and hyperparameter robustness of KDS-EL.

4.1.1. High-dimensional benchmark datasets

To validate the optimization performance and feature reduction capability of KDS-EL across diverse problem domains, a total of 20 benchmark datasets widely adopted in the evolutionary feature selection literature [13, 9, 12] are selected. As cataloged in Table 2, the suite spans two distinct operational regimes: 15 biomedical microarrays characterized by severe sample scarcity where available observations $N_{\text{samp}} \leq 173$ are vastly outnumbered by candidate features, alongside five text, image, speech, and synthetic benchmarks providing over 1,400 training observations. Such diverse data distributions and dimensional complexities effectively evaluate the combinatorial search capability and feature reduction efficacy of feature selection algorithms across contrasting sample-to-dimension regimes. The detailed characteristics of the investigated datasets are summarized in Table 2.

To guarantee experimental reproducibility and methodological hygiene, data preparation adheres to a standardized protocol across all evaluated domains. The 20 benchmark datasets are acquired from established archives, including the Kent Ridge Biomedical Data Set repository and the Arizona State University Feature Selection repository, as complete numerical feature matrices without missing values, obviating the need for statistical imputation. All comparison methods used the same source-provided numerical representation of each dataset. No additional dataset-level standardization, resampling, or imputation was performed. Where preprocessing had already been performed by the original data provider, the corresponding processed representation was retained without further transformation. Natural class distributions are preserved intact: no synthetic oversampling, random undersampling, or artificial balancing heuristics are introduced, preventing synthetic distribution artifacts. To address intrinsic class imbalance across cross-validation splits, five-fold stratified partitioning is strictly enforced, preserving identical category proportions across training and validation partitions under a synchronized seed.

4.1.2. Comparison algorithms

We benchmark KDS-EL against nine representative single-objective competitor algorithms across two primary categories, as categorized in Table 1. The first category comprises established evolutionary and swarm competitors, including FWPSO [9], PSO-CSM [12], SaWDE [10], IGPSO [11], and BPSO [8]. The second category encompasses contemporary metaheuristics, including MEL [13], DENCA-SO [14], bIRIME [15], and BinCOA [16], configured under parameters in Table 3. Regarding IGPSO [11], while the original framework estimates importance via a Multi-Layer Perceptron (MLP) on large-sample tasks, training neural surrogates under severe sample scarcity where observations $N_{\text{samp}} \ll 100$ introduces acute overfitting and parameter under-determination. This study therefore evaluates an adapted ANOVA-guided implementation variant: a parameter-free two-stage ANOVA F-statistic permutation contrast estimates feature relevance, while downstream importance-guided velocity updating, parameter modulation $(w_j, c_{1,j}, c_{2,j})$, and biased initialization follow Xue and Zhang [11] with strict fidelity. Crucially, this represents a domain-adapted implementation variant designed for sample-scarce settings rather than an exact replication of the MLP pipeline, assessing the importance-guided search principle transparently without surrogate training artifacts. For conciseness and visual consistency across tables and figures, this sample-adapted variant is designated as IGPSO throughout the empirical benchmarks. Following standard conventions in evolutionary feature selection benchmarking [13, 9, 12], all comparative evaluations operate under a synchronized population-generation cutoff protocol ($N = 20, T = 100$) rather than a strictly enforced identical fitness evaluation budget. As cataloged in Table 1 and analyzed in Section 4.2.6, constituent search heuristics across competitor paradigms induce structural variations in generational candidate evaluations; maintaining synchronized population sizes and generational cutoffs reflects established benchmarking practice in the literature while providing an operationally uniform comparison standard. While other contemporaneous evolutionary methods, such as NNS-DFPG-CSA [18], EMSWOA [17], and MPDE-MIR [36], explore specialized knowledge-guided mechanisms across alternative heuristic backbones, our empirical benchmark suite curates nine representative single-objective competitors spanning foundational swarm heuristics, contemporary state-of-the-art wrappers, and the direct architectural precursor MEL [13]. This benchmark collection ensures a focused, methodologically rigorous appraisal of knowledge guidance, dynamic weighting, and elite refinement under identical experimental conditions and scalarized fitness formulations.

4.1.3. Evaluation metrics and statistical test protocols

Algorithm performance is evaluated across three primary quantitative criteria: internal benchmark cross-validation accuracy with standard deviation, the number of selected features alongside the feature selection ratio, and computational runtime in seconds. A 3-nearest neighbor (3-NN) classifier with 5-fold

stratified cross-validation serves as the wrapper evaluation model across all benchmark datasets, ensuring parameter-free reproducibility and high sensitivity to candidate feature subset discriminability. Classification accuracy serves as the primary optimization surrogate criterion to maintain direct structural alignment with the wrapper error objective in Eq. (2) and ensure rigorous comparability with peer evolutionary literature. Furthermore, across the 17 binary and three multi-class benchmarks in Table 2, class distributions remain balanced, rendering accuracy an objective-aligned and computationally efficient surrogate for combinatorial search. Following the standardized benchmarking protocol established in top-tier high-dimensional evolutionary feature selection literature [13, 9, 12], each benchmark problem is evaluated as a self-contained combinatorial optimization instance where internal cross-validation accuracy guides evolutionary search trajectories under a synchronized population size of $N = 20$ and generation cutoff of $T = 100$ across all competing methods.

To ensure reproducibility, all algorithms are implemented in Python 3.12 utilizing NumPy, scikit-learn, and SciPy, and evaluated under a standardized single-threaded setting on an Intel Core i5-9500 CPU @ 3.00 GHz with 16 GB DDR4 RAM running 64-bit Windows 10 (detailed specifications in Supplementary Section S5). All 10 independent stochastic runs operate under synchronized population size $N = 20$ and generation cutoff $T = 100$ using the parameters in Table 3. Following the benchmark protocol adopted by MEL [13], each dataset was treated as a self-contained combinatorial optimization instance. The mutual-information vector was computed once from the complete instance and served as fixed problem-level heuristic knowledge throughout the evolutionary search. The internal five-fold stratified cross-validation procedure was used consistently as the fitness evaluation surrogate for all candidate subsets. Consequently, the reported results characterize comparative optimization performance under this benchmark protocol rather than unbiased predictive performance on an unseen external cohort.

Furthermore, to determine whether the empirical advantages of KDS-EL over competing paradigms are statistically significant rather than stochastic artifacts, we adopt a two-tier non-parametric statistical evaluation framework. At the individual benchmark dataset level, the two-sided Mann–Whitney U test at the $\alpha = 0.05$ significance level evaluates performance differences across the 10 independent stochastic runs. Across the multi-problem benchmark suite, we strictly adhere to the non-parametric statistical testing framework recommended by Demšar [41], where each problem is represented by its mean classification performance across the 10 independent runs. Specifically, the non-parametric Friedman test [42] establishes omnibus algorithm rankings across the benchmark suite, pairwise Wilcoxon signed-rank tests assess directional superiority between KDS-EL and each competitor method, and the Holm–Bonferroni step-down procedure [43] rigorously controls the family-wise error rate across multiple comparisons. Statistical hypothesis testing is conducted primarily on classification accuracy due to its fundamental stochastic distribution across candidate evaluations, whereas feature subset compactness and computational efficiency are concurrently appraised through quantitative cardinality compression ratios and empirical speedup factors under multi-criteria analysis.

4.2. Experimental analysis

This subsection presents a comprehensive empirical evaluation of KDS-EL across eight complementary analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects.

4.2.1. Comparison of classification accuracy

The KDS-EL framework was systematically evaluated alongside nine competitor algorithms and two intermediate configurations across diverse domains. Table 4 summarizes mean classification accuracies across 10 independent runs, with best values in boldface and second-best underlined; granular dispersion profiles (Mean ± Std) are cataloged in Supplementary Table S11.

Across the benchmark suite, KDS-EL achieves the highest overall mean internal benchmark cross-validation accuracy of 0.9545, securing competitive overall performance in multi-dataset aggregate evaluations. On ultra-high-dimensional microarrays characterized by severe sample sparsity, such as *ALL4*, *Colon*, and *T1D*, KDS-EL establishes pronounced discriminative advantages over peer methods. Compared with established single-objective evolutionary algorithms, including FWPSO [9], PSO-CSM [12], SaWDE [10], IGPSO [11], and BPSO [8], KDS-EL substantially elevates internal surrogate accuracy, confirming that closed-loop knowledge transfer and dynamic weight scheduling effectively mitigate random exploratory drift. Furthermore, against contemporary competitor methods published between 2023 and 2024, including BinCOA [16] and DENCA-SO [14], KDS-EL demonstrates consistent surrogate classification advantages. Compared with the original MEL algorithm [13], KDS-EL achieves an accuracy improvement of 4.52 percentage points, reaching 0.9545 compared with 0.9093, demonstrating that prior-guided initialization and knowledge-guided elite refinement fundamentally enhance localized exploitation in complex combinatorial feature landscapes.

Table 2: Characteristics of the 20 benchmark datasets.

No.	Dataset	Samples (n)	Features (D)	Classes (C)	Domain
1	Adenoma	36	7,457	2	Gene expression
2	ALL_AML	72	7,129	2	Gene expression
3	ALL1	128	12,625	2	Gene expression
4	ALL4	93	12,625	2	Gene expression
5	Colon	62	2,000	2	Gene expression
6	DLBCL	77	7,129	2	Gene expression
7	Gastric	65	22,645	2	Gene expression
8	Gastric1	144	22,283	2	Gene expression
9	Gastric2	124	22,283	2	Gene expression
10	Leukaemia	72	7,129	2	Gene expression
11	Lymphoma	45	4,026	2	Gene expression
12	Myeloma	173	12,625	2	Gene expression
13	Prostate	102	12,625	2	Gene expression
14	SRBCT	83	2,308	4	Gene expression
15	T1D	101	54,675	2	Gene expression
16	BASEHOCK	1,993	4,862	2	Text
17	PCMAC	1,943	3,289	2	Text
18	COIL20	1,440	1,024	20	Image
19	ISOLET	1,560	617	26	Speech
20	madelon	2,600	500	2	Synthetic

Table 3: Parameter configurations for all evaluated algorithms.

Algorithm / Component	Parameter setting
Common experimental parameters:	
Population size N	20
Maximum generations $T_{\max}$	100
Classifier	3-NN with 5-fold stratified CV
Binarization threshold θ	0.60
Independent runs	10
Proposed KDS-EL framework:	
Subpopulations	$N_1 = 10,\ N_2 = 10$
Inertia weight ω	0.90
Acceleration coefficients c_1, c_2, c_3	2.0, 2.0, 2.0
Knowledge decay factor γ	0.99
Perturbation fraction ϕ	0.01
Elite pool size k_e	3
Random exploration rate p_{rand}	0.10
MI initialization ratio ρ	0.50
Gaussian perturbation scale σ	0.10
Stagnation threshold τ	20 generations
Phase split ratio κ	0.70
Phase-1 weights $\mathbf{w}_1$	[0.70, 0.30]
Phase-2 weights $\mathbf{w}_2$	[0.95, 0.05]
Intermediate configurations:	
KDS-EL w/o TPS	Static weights $\mathbf{w}_s$ = [0.90, 0.10]
KDS-EL w/o TPS&KER	Without KER and TPS, $\mathbf{w}_s$ = [0.90, 0.10]
MEL [13]	Uniform init, $\mathbf{w}_s$ = [0.90, 0.10]
Competitor algorithms:	
FWPSO [9]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ \alpha_{\text{sa}} = 0.95$
PSO-CSM [12]	$\omega = 0.7298,\ c_1 = c_2 = 1.49618,\ p_m = 0.05$
SaWDE [10]	$F \in [0.1, 0.9],\ CR \in [0.1, 0.9]$
IGPSO [11]	$\omega_j \in [0.1, 0.9],\ c_{1,j} = c_{2,j} \in [0.1, 2.0]$, ANOVA contrast
BPSO [8]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ V_{\max} = 6.0$
DENCA-SO [14]	$F = 0.5,\ CR = 0.2,\ W = 0.05$, NCFS $\lambda = 0.01$
bIRIME [15]	S-transfer, CMS mutation, restart limit $L = 5$
BinCOA [16]	ROBL, Sigmoid transfer, crisscross (HC/VC)

Table 4: Comparison of classification accuracy across benchmark datasets.

Dataset	KDS-EL	w/o TPS	w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO†	DENCA-SO	BPSO
ALL1	**1.0000**	**1.0000** ≈	**1.0000** ≈	**1.0000** ≈	**1.0000** ≈	**1.0000** ≈	0.9995 +	**1.0000** ≈	**1.0000** ≈	**1.0000** ≈	**1.0000** ≈	**1.0000** ≈
ALL4	**0.9480**	0.9367 ≈	0.9356 +	0.8656 +	0.9140 +	0.8580 +	0.8308 +	0.7634 +	0.8250 +	0.7918 +	0.7043 +	0.7846 +
ALL_AML	**1.0000**	**1.0000** ≈	0.9947 ≈	0.9857 +	**1.0000** ≈	0.9914 +	0.9868 +	0.9861 +	0.9857 +	0.9857 +	0.9778 +	0.9871 +
Adenoma	**1.0000**	**1.0000** ≈	0.9971 ≈	0.9714 +	**1.0000** ≈	0.9800 +	0.9767 +	0.9722 +	0.9714 +	0.9714 +	0.9611 +	0.9714 +
BASEHOCK	0.9045	0.8933 +	0.8627 +	0.8802 +	0.9082 ≈	0.9045 ≈	0.8739 +	0.8897 +	**0.9170** ≈	0.8743 +	0.8610 +	0.8994 ≈
COIL20	**1.0000**	0.9976 +	0.9955 +	0.9944 +	0.9991 +	0.9980 +	0.9939 +	0.9953 +	0.9988 +	0.9951 +	0.9942 +	0.9967 +
Colon	0.9452	0.9294 +	0.9182 +	0.8777 +	**0.9468** ≈	0.9268 +	0.8914 +	0.8694 +	0.9086 +	0.8733 +	0.8048 +	0.8885 +
DLBCL	0.9999	**1.0000** ≈	0.9987 ≈	0.9859 +	**1.0000** ≈	0.9935 ≈	0.9966 +	0.9883 +	0.9924 +	0.9884 +	0.9675 +	0.9899 +
Gastric	**0.9763**	0.9646 +	0.9723 ≈	0.9015 +	0.9569 +	0.9154 +	0.8886 +	0.8231 +	0.8385 +	0.8646 +	0.7769 +	0.8369 +
Gastric1	**0.9779**	0.9719 +	0.9636 +	0.9602 +	0.9722 +	0.9623 +	0.9588 +	0.9583 +	0.9623 +	0.9588 +	0.9500 +	0.9581 +
Gastric2	**0.9982**	0.9936 +	0.9880 +	0.9822 +	0.9952 ≈	0.9887 +	0.9830 +	0.9823 +	0.9815 +	0.9837 +	0.9750 +	0.9815 +
ISOLET	0.8880	0.8955 −	0.8801 +	0.8783 +	0.9112 −	0.9000 −	0.8775 ≈	0.8763 +	**0.9164** −	0.8648 +	0.8760 +	0.8942 −
Leukaemia	**1.0000**	**1.0000** ≈	0.9960 ≈	0.9857 +	**1.0000** ≈	0.9957 ≈	0.9895 +	0.9861 +	0.9857 +	0.9871 +	0.9764 +	0.9857 +
Lymphoma	**1.0000**	**1.0000** ≈	0.9867 +	0.8978 +	0.9978 ≈	0.9422 +	0.9263 +	0.8889 +	0.9044 +	0.8978 +	0.8400 +	0.8933 +
Myeloma	**0.9083**	0.8889 +	0.8710 +	0.8466 +	0.8884 +	0.8722 +	0.8552 +	0.8387 +	0.8757 +	0.8367 +	0.8012 +	0.8536 +
PCMAC	**0.8592**	0.8360 +	0.8000 +	0.8085 +	0.8410 +	0.8301 +	0.8038 +	0.8161 +	0.8498 ≈	0.7985 +	0.7601 +	0.8310 +
Prostate	**0.9670**	0.9552 +	0.9600 +	0.8732 +	0.9353 +	0.8930 +	0.8715 +	0.8686 +	0.8940 +	0.8603 +	0.8235 +	0.8863 +
SRBCT	**1.0000**	**1.0000** ≈	**1.0000** ≈	0.9682 +	**1.0000** ≈	0.9865 +	0.9837 +	0.9663 +	0.9976 ≈	0.9709 +	0.9325 +	0.9953 ≈
T1D	**0.8449**	0.8043 +	0.8054 +	0.7558 +	0.7752 +	0.7446 +	0.7204 +	0.7178 +	0.7296 +	0.7060 +	0.6723 +	0.7285 +
madelon	**0.8731**	0.8567 ≈	0.7710 +	0.7667 +	0.8677 ≈	0.8008 +	0.7621 +	0.7598 +	0.8255 +	0.7599 +	0.7695 +	0.7882 +
Average	0.9545	0.9462	0.9348	0.9093	0.9454	0.9242	0.9085	0.8973	0.9180	0.8985	0.8712	0.9075
+/ ≈ /−	—	11/8/1	13/7/0	19/1/0	8/11/1	15/4/1	19/1/0	19/1/0	15/3/2	19/1/0	19/1/0	17/2/1

Note: Bold: best; underline: second-best. +/ ≈ /−: statistically superior, equivalent, or inferior to KDS-EL under pairwise two-sided Mann–Whitney U tests ($\alpha = 0.05$). †Denotes the sample-adapted ANOVA implementation variant detailed in Section 4.1.

A granular examination of per-dataset performance in Table 4 reveals nuanced trade-off dynamics between KDS-EL and the strongest contemporary competitor, bIRIME [15]. Head-to-head on raw mean classification accuracy, KDS-EL records 11 wins, 5 ties, and 4 losses against bIRIME across the 20 benchmark datasets, which translates into 8 statistically significant wins, 11 ties, and only 1 loss under the two-sided Mann–Whitney U test in Table 6. Among the four benchmark instances where bIRIME records a higher raw mean, three instances, namely *BASEHOCK*, *Colon*, and *DLBCL*, exhibit marginal accuracy differentials below 0.0038 that are statistically indistinguishable under the Mann–Whitney U test, with both optimizers reaching near-saturation on *DLBCL* at 0.9999 and 1.0000, respectively. bIRIME establishes a statistically significant advantage on only a single benchmark instance, *ISOLET*, where it records 0.9112 compared with 0.8880. Conversely, on challenging non-saturated domains such as *ALL4*, *Prostate*, and *T1D*, KDS-EL achieves pronounced accuracy gains over bIRIME. Furthermore, as documented in Table 5, bIRIME maintains its competitive accuracy at the expense of substantial feature bloat, retaining an average of 407.7 features across the benchmark suite compared with 87.8 features isolated by KDS-EL, which corresponds to a 78.5% reduction in feature cardinality. Additionally, as analyzed in Section 4.2.6, bIRIME incurs a steep wall-clock runtime penalty averaging 326.2 seconds due to unguided multi-candidate evaluations, running over six times slower than KDS-EL. Consequently, while bIRIME maintains competitive accuracy parity on specific benchmark instances, KDS-EL establishes a favorable multi-criteria profile: delivering competitive classification accuracy alongside approximately 4.6 times fewer features and over a six-fold reduction in computational runtime.

4.2.2. Comparison of the number of selected features

Table 5 reports the selected feature subset sizes across the evaluated competitor methods and intermediate configurations. Across the benchmark suite, KDS-EL achieves marked feature subset compactness, isolating an average of 87.8 features and compressing ultra-high-dimensional spaces by approximately 98.0%. This dimensionality reduction is particularly prominent on biomedical microarrays, such as *ALL1*, *Colon*, *Gastric1*, and *Prostate*, where compact subsets of fewer than 40 dimensions preserve high classification accuracy.

In contrast, conventional evolutionary feature selection algorithms exhibit severe dimensional redundancy. Even the most competitive peer wrapper, FWPSO [9], retains an average of over 1,000 features, while contemporary meta-heuristics such as bIRIME [15] preserve over 400 features due to the absence of targeted importance decay guidance. Most critically, canonical swarm heuristics, including BinCOA [16], DENCA-SO [14], and standard BPSO [8], suffer from severe dimensional stagnation, selecting over 5,000 redundant features across high-dimensional landscapes. This degradation occurs because unguided binarization mechanisms inherently center active dimension probabilities around 0.50 when proactive pruning pressure is absent. By contrast, KDS-EL combines continuous mutual information prior seeding with aggressive early-stage feature penalization in Two-Phase Weight Scheduling, actively steering candidate trajectories into ultra-sparse, discriminative subspaces. This dimensional compression remains consistent across moderate-sample benchmarks such as *BASEHOCK* and *ISOLET* where observational counts reach thousands: while competitors preserve 1,113 to 2,445 and 133 to 374 features to resolve dense boundaries, KDS-EL isolates merely 134 and 60 dimensions, respectively, maintaining competitive classification accuracy on the evaluation surrogate alongside a substantial reduction in feature redundancy.

Evaluating the intermediate algorithmic configurations demonstrates how each constituent mechanism systematically compresses feature redundancy. Across the benchmark suite, feature cardinality contracts hierarchically from over 2,700 dimensions in the pre-improvement configuration down to fewer than 300 dimensions through mutual information prior seeding, further halves via knowledge-guided elite refinement, and ultimately consolidates into an ultra-compact signature of 87.8 features under dynamic two-phase scheduling. This progressive reduction confirms that the proposed components operate in a complementary cascade, where domain prior seeding suppresses global noise, elite refinement purges weakly contributing dimensions, and two-phase scheduling enforces rigorous dimensional parsimony.

Crucially, while feature reduction progresses monotonically in terms of the multi-dataset average, subset cardinality on individual problem instances reflects a deliberate, domain-specific trade-off governed by two-phase scheduling dynamics. On several challenging biomedical microarrays, such as *Adenoma*, *Lymphoma*, and *SRBCT*, the complete KDS-EL framework retains slightly more features than the static-weight variant without TPS. This behavior directly manifests the underlying design rationale of the two-phase strategy: upon transitioning to Phase 2, the feature-cardinality penalty is relaxed while classification accuracy is prioritized. Rather than enforcing an inflexible sparsity constraint that risks permanently pruning synergistic feature interactions, this controlled relaxation empowers the search engine to strategically retrieve a small subset of weakly individual but strongly synergistic features that resolve ambiguous classification boundaries. Consequently, this targeted trade-off converts marginal feature retention into a statistically confirmed accuracy advantage, confirming that KDS-EL successfully harmonizes extreme feature sparsity with competitive surrogate classification accuracy.

4.2.3. Fitness value and convergence analysis

The scalar fitness function (27) formalizes the trade-off between minimizing classification error and compressing active feature ratios, quantifying exploratory efficiency and exploitation depth. To evaluate evolutionary trajectories comprehensively, Fig. 5 plots mean fitness convergence over 100 generations for KDS-EL alongside four representative wrapper methods: FWPSO [9], SaWDE [10], MEL [13], and BPSO [8]. Trajectories depict mean profiles across 10 runs to avoid visual occlusion from overlapping shaded error bands; full run-level dispersion and variances are reported via boxplots in Supplementary Fig. S6 and Table S11.

As demonstrated across the individual panels in Fig. 5, KDS-EL establishes a favorable convergence profile, achieving the lowest final fitness values across virtually all evaluated datasets. In the initial generation, mutual in-

Table 5: Comparison of selected feature subset size across benchmark datasets.

Dataset	KDS-EL	w/o TPS	w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO†	DENCA-SO	BPSO
ALL1	**39**	55	55	4248	65	612	1206	6120	3667	2501	5877	5886
ALL4	124	152	**64**	1082	393	1326	1447	6301	4581	2677	6111	6217
ALL_AML	**67**	70	473	2369	119	521	949	3419	1996	1444	3251	3266
Adenoma	66	**33**	120	2483	40	278	937	3554	2102	1608	3397	3399
BASEHOCK	**134**	208	812	1965	1721	2066	1113	2445	1912	1894	2359	2419
COIL20	84	60	348	323	**54**	90	167	454	232	255	634	407
Colon	**14**	35	105	702	113	240	284	960	621	448	845	935
DLBCL	64	**52**	615	2784	247	876	1074	3514	2155	1718	3784	3379
Gastric	100	97	**62**	3604	258	1070	2782	11157	7072	4916	10625	11004
Gastric1	**31**	217	678	7393	213	1420	2966	10937	6963	5074	10911	10661
Gastric2	**83**	140	444	7084	348	1362	3062	11041	6708	5134	10971	10627
ISOLET	**60**	100	225	250	133	172	194	298	213	195	374	272
Leukaemia	66	**47**	488	2409	132	534	934	3406	2026	1440	3218	3246
Lymphoma	45	**23**	50	1404	137	304	524	1934	1204	880	1765	1832
Myeloma	250	365	**175**	2137	1160	1863	1847	6278	4543	2859	6596	6259
PCMAC	**91**	329	637	1339	928	1209	651	1621	1292	1137	1590	1635
Prostate	26	**23**	27	5058	418	1898	1541	6282	4229	2729	5837	6211
SRBCT	64	**36**	129	906	172	431	366	1136	739	515	1056	1083
T1D	330	366	**174**	6715	1459	4984	9047	27318	18009	13170	27145	26971
madelon	**18**	26	130	190	46	122	95	240	162	171	122	221
Average	**87.8**	121.7	290.5	2722.2	407.7	1068.9	1559.4	5420.7	3521.1	2538.2	5323.4	5296.5
Comp. Ratio	**98.0%**	97.2%	91.2%	68.4%	92.7%	86.0%	84.5%	51.3%	68.2%	75.3%	52.3%	53.0%

Note: Bold: most compact; underline: second-most compact. Per-dataset entries report mean feature subset sizes rounded to the nearest integer; aggregate values are computed directly from unrounded run-level evaluation results. †Denotes the sample-adapted ANOVA implementation variant detailed in Section 4.1.

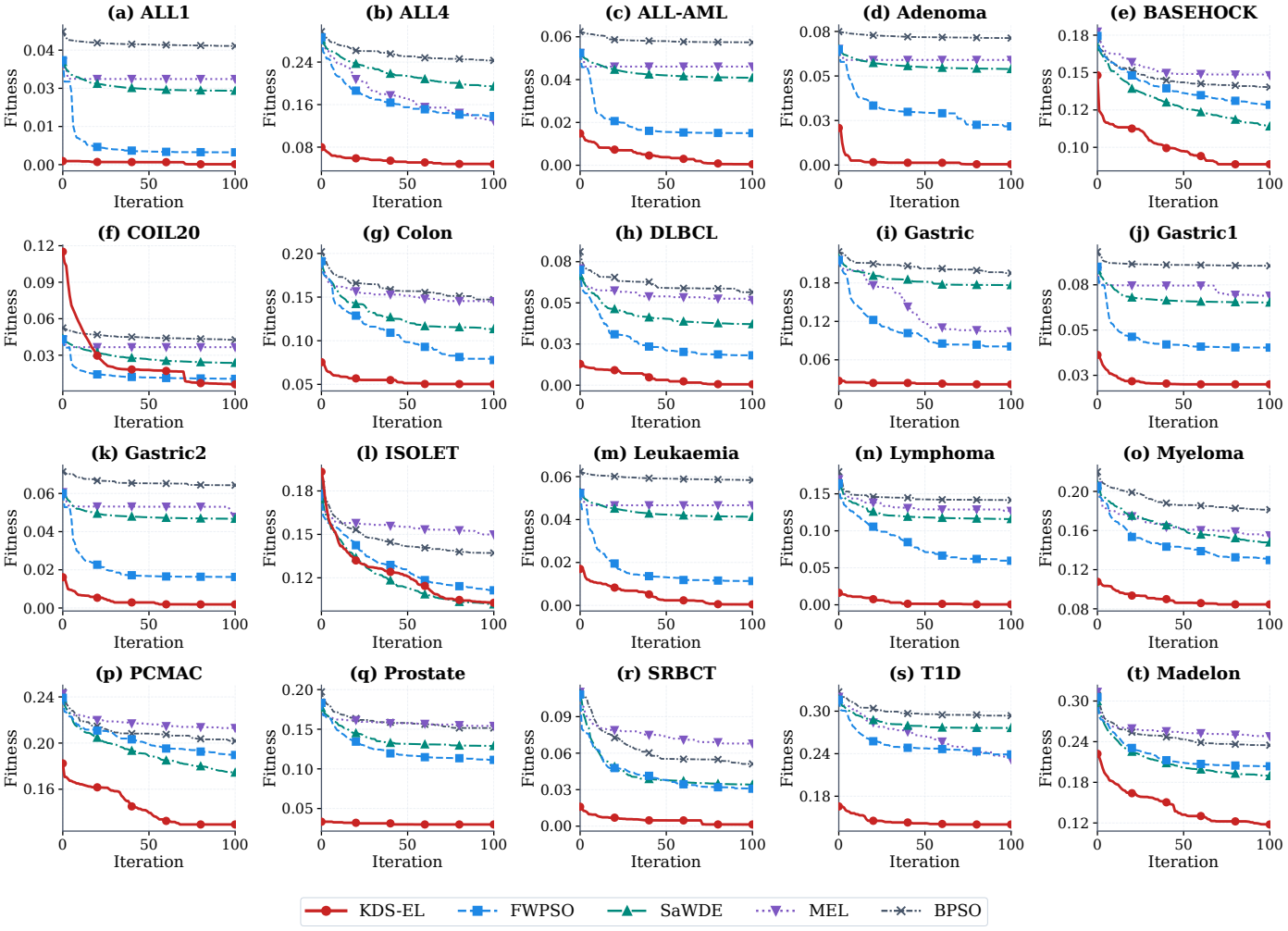


Figure 5: Convergence trajectories of mean objective fitness across all 20 benchmark datasets.

formation continuous prior seeding directly places the swarm within compact, discriminative subspaces. Consequently, KDS-EL commences search from an initial reference fitness substantially lower than those of competing swarms, which incur severe initial penalties due to unguided random bit assignments that indiscriminately activate thousands of noisy features. During early iterations, the aggressive feature-size penalization in Two-Phase Weight Scheduling accelerates dimensional pruning, producing a steep, monotonic fitness descent across both biomedical microarrays, including *ALL4*, *Colon*, and *Gastric*, and heterogeneous vision and text benchmarks, such as *BASEHOCK* and *COIL20*, while competing methods struggle to escape dense initial configurations.

The comparative trajectories further reveal distinct search stagnation behaviors among the competing paradigms. Standard BPSO [8] exhibits erratic, sluggish progress across high-dimensional landscapes, frequently stagnating after generation 20 due to uniform velocity clipping and the absence of do-

Table 6: Pairwise statistical comparison on classification accuracy.

Method	KDS-EL	w/o TPS	w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
KDS-EL	—	11/8/1	13/7/0	19/1/0	8/11/1	15/4/1	19/1/0	19/1/0	15/3/2	19/1/0	19/1/0	17/2/1
w/o TPS	1/8/11	—	11/9/0	19/1/0	3/11/6	12/7/1	20/0/0	18/2/0	14/2/4	19/1/0	19/1/0	15/5/0
w/o TPS&KER	0/7/13	0/9/11	—	15/4/1	3/7/10	7/8/5	16/3/1	15/3/2	10/5/5	16/4/0	16/4/0	14/1/5
MEL	0/1/19	0/1/19	1/4/15	—	0/2/18	0/8/12	4/12/4	4/10/6	1/10/9	4/16/0	12/8/0	1/12/7
bIRIME	1/11/8	6/11/3	10/7/3	18/2/0	—	15/5/0	20/0/0	19/1/0	14/5/1	19/1/0	19/1/0	19/1/0
FWPSO	1/4/15	1/7/12	5/8/7	12/8/0	0/5/15	—	18/2/0	14/6/0	8/6/6	16/4/0	18/2/0	12/8/0
PSO-CSM	0/1/19	0/0/20	1/3/16	4/12/4	0/0/20	0/2/18	—	10/6/4	4/4/12	9/8/3	11/8/1	5/5/10
BinCOA	0/1/19	0/2/18	2/3/15	6/10/4	0/1/19	0/6/14	4/6/10	—	3/4/13	6/8/6	17/3/0	3/3/14
SaWDE	2/3/15	4/2/14	5/5/10	9/10/1	1/5/14	6/6/8	12/4/4	13/4/3	—	12/7/1	15/5/0	10/10/0
IGPSO	0/1/19	0/1/19	0/4/16	0/16/4	0/1/19	0/4/16	3/8/9	6/8/6	1/7/12	—	12/7/1	1/9/10
DENCA-SO	0/1/19	0/1/19	0/4/16	0/8/12	0/1/19	0/2/18	1/8/11	0/3/17	0/5/15	1/7/12	—	0/6/14
BPSO	1/2/17	0/5/15	5/1/14	7/12/1	0/1/19	0/8/12	10/5/5	14/3/3	0/10/10	10/9/1	14/6/0	—

Note: $W/T/L$: counts of benchmark datasets with statistically superior (W), equivalent (T), or inferior (L) classification accuracy under pairwise two-sided Mann–Whitney U tests ($\alpha = 0.05$).

main guidance. While FWPSO [9] and SaWDE [10] achieve progressive fitness improvements, their trajectories plateau prematurely between generations 40 and 60 because memoryless stochastic perturbations discard iteration feedback and fail to reinforce synergistic feature combinations. Similarly, canonical dual-swarm search [13] suffers from an extended exploratory plateau during early generations before locating viable descent directions. In contrast, KDS-EL sustains continuous fitness reduction throughout the entire 100-generation timeline. By dynamically accumulating historical feature-importance feedback within the importance vector $\mathbf{W}$ with exponential decay γ and steering elite perturbations toward high-utility candidates via the KER strategy, KDS-EL continually escapes local attractors, securing an order-of-magnitude fitness reduction on ultra-high-dimensional benchmarks such as *ALL1*, *DLBCL*, *Leukaemia*, *Lymphoma*, and *T1D*.

4.2.4. Multi-tier non-parametric statistical significance analysis

To evaluate whether the empirical performance advantages of KDS-EL are statistically meaningful rather than stochastic artifacts, we evaluate non-parametric significance across both analytical tiers defined in Section 4.1. At the single-dataset level, Table 6 reports the pairwise comparison matrix based on two-sided Mann–Whitney U tests at significance level $\alpha = 0.05$ across the 10 independent stochastic runs on each benchmark problem. Each entry (i, j) records the $W/T/L$ counts denoting the number of benchmark datasets where method i attained statistically superior, equivalent, or inferior accuracy compared with method j. Across the comparison suite, KDS-EL demonstrates pronounced surrogate classification advantages over established single-objective metaheuristics, such as FWPSO [9], SaWDE [10], PSO-CSM [12], IGPSO [11], and standard BPSO [8], as well as contemporary competitors including BinCOA [16] and DENCA-SO [14]. Notably, against MEL [13], KDS-EL achieves 19 statistically significant wins and 1 tie without conceding a single loss. Against the leading competitor bIRIME, KDS-EL exhibits localized classification parity, recording 8 wins, 11 ties, and 1 loss under Mann–Whitney U tests, with statistical ties concentrating on saturated ceiling-effect domains where both optimizers achieve near-perfect classification while KDS-EL isolates far sparser feature subsets. Furthermore, significant superiority over the intermediate configuration without TPS, documented by 11 wins, 8 ties, and only 1 loss, confirms that dynamic two-phase scheduling delivers consistent accuracy gains over static weighting.

Complementing the single-dataset 10-run comparisons, the second tier evaluates multi-problem significance following Demšar's methodology [41], where the analysis unit strictly corresponds to the $N = 20$ benchmark instances represented by their 10-run mean classification accuracies rather than composite scalarized fitness, preventing sample size pseudoreplication. Table 7 summarizes the post-hoc Holm–Bonferroni step-down analysis [43] with KDS-EL as control method, reporting within-dataset Mann–Whitney $W/T/L$ counts mirrored from Table 6 alongside paired Wilcoxon signed-rank test statistics across the 20 dataset means. Specifically, competitor methods are sorted by ascending unadjusted p-values p_{raw} of two-sided Wilcoxon signed-rank tests on paired mean accuracy differentials ΔAcc under standard zero-difference exclusion with effective sample size $n_{\text{eff}} \leq 20$, with W denoting the negative rank sum and $r_{\text{rb}} \in [-1, 1]$ the matched-pairs rank-biserial effect size. As shown in Table 7, KDS-EL establishes pronounced positive effect sizes across all competitor methods with $r_{\text{rb}} \geq +0.633$ and intermediate configurations with $r_{\text{rb}} \geq +0.886$. Notably, the fourfold concordance in $p_{\text{raw}} = 1.32 \times 10^{-4}$ for MEL, BinCOA, IGPSO, and DENCA-SO reflects identical rank distributions: each exhibits 19 losses and 1 tie against KDS-EL, observed on the ceiling benchmark *ALL1*, yielding $n_{\text{eff}} = 19$, $W = 0.0$, and $r_{\text{rb}} = +1.000$. Against bIRIME, statistical rejection is attained at the family-wise boundary with $p_{\text{raw}} = 0.0309 \leq \alpha_{\text{Holm}} = 0.0500$ and $r_{\text{rb}} = +0.633$, establishing an overall accuracy advantage accompanied by descriptive compactness and runtime gains, including an approximately 4.6-fold feature reduction from 407.7 to 87.8 dimensions alongside over a six-fold computational speedup.

To establish an omnibus global algorithmic ranking across the benchmark suite, the non-parametric Friedman test [42] is conducted strictly on 10-run mean classification accuracy rather than composite scalarized fitness, evaluated directly from the unrounded floating-point data matrix to prevent artificial ties induced by decimal display truncation. The omnibus test rejects the null hypothesis of equivalent classification performance across the 20 benchmark datasets, yielding $\chi_F^2 = 155.69$ (df = 11, $p < 0.001$) alongside Iman–Davenport $F_F = 46.00$ ($p < 0.001$), formally validating post-hoc pairwise comparisons. Where the optimizer achieving the highest mean accuracy on each benchmark instance receives rank 1, Fig. 6 illustrates the resulting average ranks: KDS-EL attains the top average Friedman rank of 1.90 across the entire comparison suite, outperforming leading peer algorithms such as bIRIME with an average rank of 2.60 as well as its own intermediate configurations. In contrast, standard evolutionary algorithms and unguided wrappers occupy noticeably inferior positions, reflecting their vulnerability to premature stagnation in ultra-high dimensions. This Friedman ranking trajectory confirms that KDS-EL achieves superior accuracy-based omnibus rankings alongside its multi-criteria dimensional compactness across heterogeneous problem distributions.

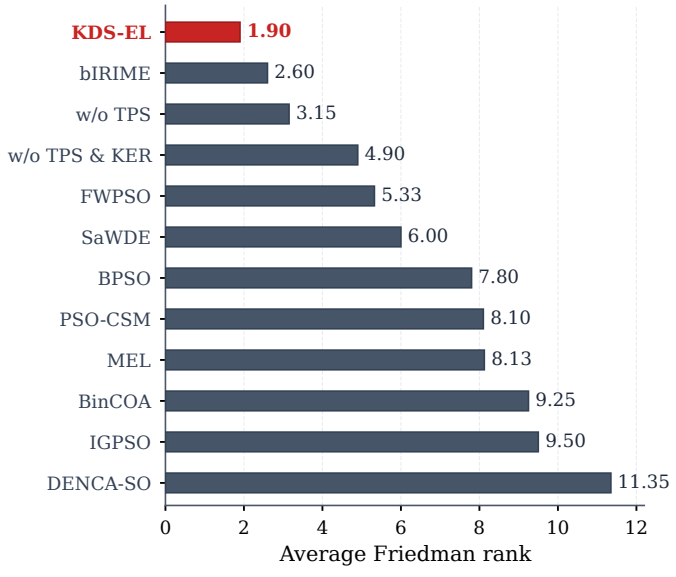


Figure 6: Average Friedman rankings on classification accuracy across benchmark datasets.

4.2.5. Component contribution and progressive mechanism analysis

To evaluate constituent mechanism contributions without confounding variables, comparative experiments assess a four-stage progression from pre-improvement MEL to KDS-EL. Table 8 summarizes the comparative performance profiles reporting Mean ± Std, with best accuracies in boldface and most compact subsets underlined. Granular metrics and standard deviations for all intermediate configurations are documented in Supplementary Tables S4 and S11.

A critical insight emerging from the empirical progression, visualized

Table 7: Holm–Bonferroni step-down procedure on classification accuracy with KDS-EL as control method.

Rank	Competitor Method	ΔAcc	$W/T/L$	n_{eff}	W	r_{rb}	p_{raw}	p_{adj}	α_{Holm}
1	PSO-CSM	+0.0460	19/1/0	20	0.0	+1.000	1.91e-06	2.10e-05	0.0045
2	MEL	+0.0452	19/1/0	19	0.0	+1.000	1.32e-04	1.32e-03	0.0050
3	BinCOA	+0.0572	19/1/0	19	0.0	+1.000	1.32e-04	1.32e-03	0.0056
4	IGPSO	+0.0561	19/1/0	19	0.0	+1.000	1.32e-04	1.32e-03	0.0063
5	DENCA-SO	+0.0833	19/1/0	19	0.0	+1.000	1.32e-04	1.32e-03	0.0071
6	w/o TPS&KER	+0.0197	13/7/0	18	0.0	+1.000	1.96e-04	1.32e-03	0.0083
7	BPSO	+0.0470	17/2/1	19	4.0	+0.958	2.50e-04	1.32e-03	0.0100
8	FWPSO	+0.0303	15/4/1	19	7.0	+0.926	3.98e-04	1.59e-03	0.0125
9	SaWDE	+0.0365	15/3/2	19	15.0	+0.842	1.28e-03	3.85e-03	0.0167
10	w/o TPS	+0.0083	11/8/1	14	6.0	+0.886	3.51e-03	7.02e-03	0.0250
11	bIRIME	+0.0091	8/11/1	15	22.0	+0.633	3.09e-02	3.09e-02	0.0500

Note: W: Wilcoxon signed-rank statistic across 20 dataset means; r_{rb}: matched-pairs rank-biserial effect size; p_{adj}: Holm-adjusted p-value.

Table 8: Performance comparison between original MEL and KDS-EL.

Dataset	MEL		Proposed KDS-EL	
	Acc ± Std	Number	Acc ± Std	Number
ALL1	1.0000±0.0000	4,248	**1.0000±0.0000**	39
ALL4	0.8656±0.0558	1,082	**0.9480±0.0068**	124
ALL_AML	0.9857±0.0000	2,369	**1.0000±0.0000**	67
Adenoma	0.9714±0.0000	2,483	**1.0000±0.0000**	66
BASEHOCK	0.8802±0.0079	1,965	**0.9045±0.0117**	134
COIL20	0.9944±0.0009	323	**1.0000±0.0000**	84
Colon	0.8777±0.0193	702	**0.9452±0.0082**	14
DLBCL	0.9859±0.0074	2,784	**0.9999±0.0002**	64
Gastric	0.9015±0.0649	3,604	**0.9763±0.0038**	100
Gastric1	0.9602±0.0047	7,393	**0.9779±0.0002**	31
Gastric2	0.9822±0.0075	7,084	**0.9982±0.0030**	83
ISOLET	0.8783±0.0066	250	**0.8880±0.0084**	60
Leukaemia	0.9857±0.0000	2,409	**1.0000±0.0000**	66
Lymphoma	0.8978±0.0155	1,404	**1.0000±0.0000**	45
Myeloma	0.8466±0.0092	2,137	**0.9083±0.0112**	250
PCMAC	0.8085±0.0085	1,339	**0.8592±0.0126**	91
Prostate	0.8732±0.0092	5,058	**0.9670±0.0036**	26
SRBCT	0.9682±0.0107	906	**1.0000±0.0000**	64
T1D	0.7558±0.0313	6,715	**0.8449±0.0105**	330
madelon	0.7667±0.0124	190	**0.8731±0.0188**	18
Average	0.9093±0.0136	2,722.2	**0.9545±0.0049**	87.8
Δ vs. MEL			**+4.52 pp**	-96.8%

Note: Bold: higher accuracy; underline: more compact feature subset.

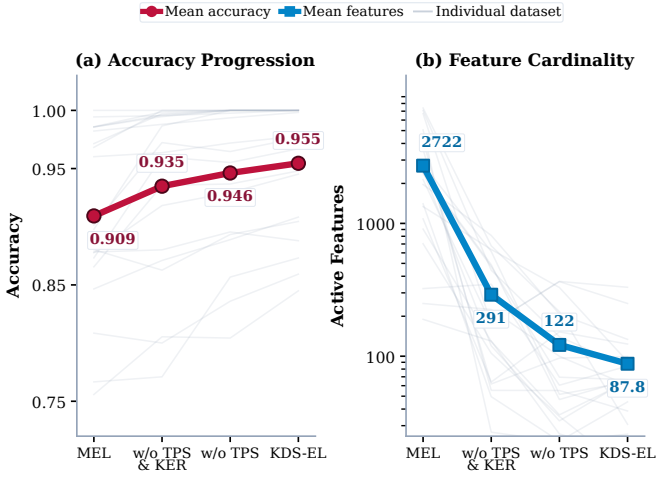


Figure 7: Accuracy and feature progression across constituent mechanisms.

through multi-dataset accuracy progression profiles in Fig. 7 and convergence curves in Fig. 8, concerns asymmetric search complexity across constituent mechanisms. Transitioning from the pre-improvement configuration to KDS-EL w/o TPS&KER yields the largest numerical shift in raw accuracy, producing a 2.55 pp elevation from 0.9093 to 0.9348 alongside a dimensional contraction from 2,722.2 to 290.5 features. Methodologically, this surge reflects the combined necessity of mutual information soft prior initialization and adaptive diversity recovery. In ultra-high-dimensional spaces, unguided random initialization in the pre-improvement configuration indiscriminately activates approximately half of all candidate dimensions, trapping particles within dense, uninformative subspaces. By continuous domain-relevance seeding, the MII mechanism immediately places the initial swarm within compact, discriminative subspaces, while the AIR strategy recovers stagnant particles and restores momentum, reducing more than 2,400 selected dimensions compared with unguided search.

Crucially, static filter-based relevance and stochastic restarts cannot navigate the combinatorial fine-tuning required in small-sample regimes: retaining 290.5 features leaves candidate subsets vulnerable to finite-sample correlation noise and the Winner's curse. The core evolutionary intelligence of KDS-EL resides in the subsequent closed-loop cascade that replaces unguided bidirectional updates with feedback-driven refinement. Specifically, the KER strategy harvests generational fitness differentials within the importance vector $\mathbf{W}$ under exponential forgetting, steering targeted perturbations toward high-utility candidate features. This closed-loop exploitation delivers an additional 1.14 pp accuracy gain while reducing retained features to 121.7 in KDS-EL w/o TPS. Finally, compared with static scalarization in KDS-EL w/o TPS, the dynamic TPS strategy temporally shifts selection pressure from coarse pruning to fine-grained discrimination, securing a further 0.83 pp gain and compressing the signature to 87.8 features. Together, the KER and TPS strategies achieve an intensive 1.97 pp accuracy breakthrough and prune over 200 additional candidate dimensions beyond prior seeding, confirming that KDS-EL establishes a cohesive closed-loop evolutionary system rather than a superficial filter pre-selector.

4.2.6. *Computational efficiency and runtime analysis*

Computational efficiency constitutes a vital evaluation dimension for evolutionary feature selection, particularly within wrapper-based metaheuristics where iterative classifier evaluations dominate wall-clock execution time. To assess real-world computational tractability, Fig. 9 visualizes the empirical trade-off distribution between classification accuracy and selected feature cardinality across the evaluated problem domains. Comprehensive per-dataset execution measurements across all evaluated algorithms are systematically documented in Supplementary Table S5.

As illustrated in Fig. 9 and detailed in Supplementary Table S5, KDS-EL establishes a highly favorable efficiency profile, achieving an omnibus average execution time of 51.3 seconds across the benchmark suite and delivering a 3.4-fold speedup factor ($S = t_{\text{MEL}}/t_{\text{KDS-EL}} = 175.0\,\text{s}/51.3\,\text{s} \approx 3.41$) over canonical dual-swarm search under the standardized single-threaded evaluation environment on an Intel Core i5-9500 CPU. While evolutionary filter paradigms such as DENCA-SO [14] exhibit rapid execution by evaluating candidate subsets through algebraic surrogate metrics rather than inductive classifiers, they do so at the cost of substantial surrogate accuracy deficits and severe feature bloat. Conversely, contemporary state-of-the-art wrapper optimizers, including bIRIME [15] and BinCOA [16], suffer from steep computational penalties that escalate dramatically on large-scale and high-dimensional problem instances. Standard evolutionary architectures such as PSO-CSM [12], SaWDE [10], and BPSO [8] likewise incur prolonged execution times because unguided stochastic updates wander extensively across dense combinatorial spaces. In contrast, KDS-EL consistently occupies the ideal high-accuracy, ultra-sparse, and low-latency region in the upper-left quadrant of Fig. 9, achieving practical scalability without compromising classification accuracy on the evaluation surrogate.

Under the empirical implementation and generation-cutoff settings of this study, KDS-EL exhibits a pronounced wall-clock advantage of 51.3 s over the 175.0 s required by MEL and 326.2 s by bIRIME. To ensure methodological rigor, we distinguish the structural differences in generational evaluation bud-

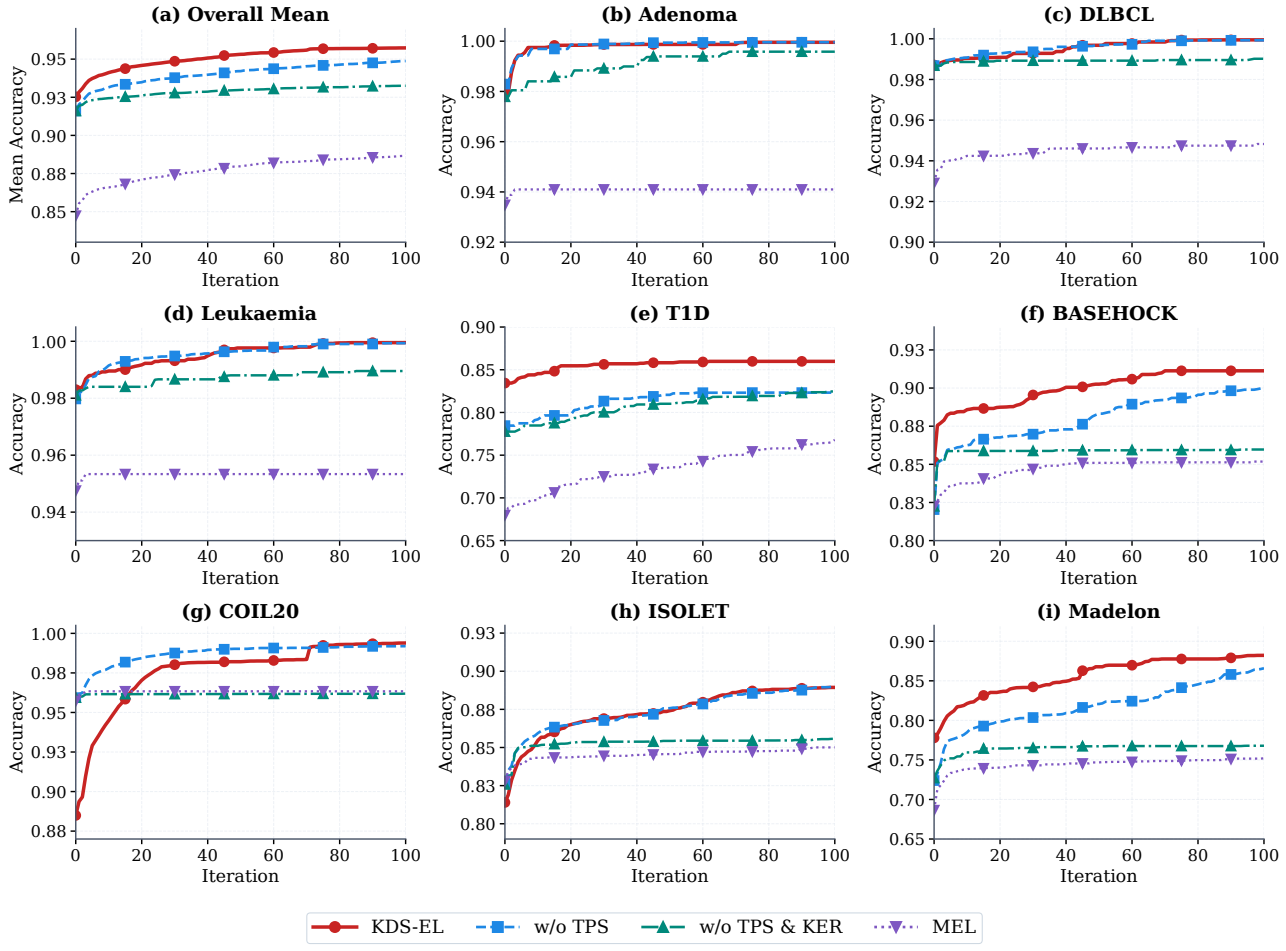


Figure 8: Convergence trajectories of classification accuracy across constituent mechanism stages.

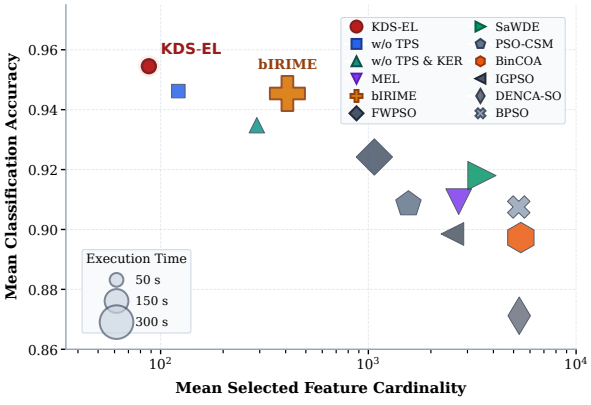


Figure 9: Trade-off distribution across accuracy, feature cardinality, and execution time.

gets across competitor paradigms under the standardized 100-generation cutoff. Standard evolutionary optimizers, including BPSO, FWPSO, SaWDE, and MEL, evaluate exactly 20 candidate individuals per generation, totaling 2,000 candidate evaluations across a complete run. In contrast, KDS-EL evaluates 20 particles alongside three elite refinements per generation, amounting to approximately 2,300 evaluations, whereas bIRIME deploys Composite Mutation Strategies that evaluate up to 80 candidate subsets per generation. Crucially, despite executing slightly more function evaluations than standard swarm optimizers, KDS-EL achieves a 3.4-fold wall-clock acceleration. This empirical efficiency gain is primarily driven by active feature compression: in wrapper feature selection, pairwise Euclidean distance computations scale linearly with the active feature cardinality. By rapidly contracting candidate subsets from thousands of raw dimensions down to an average of 87.8 features within early generations, KDS-EL substantially diminishes per-evaluation distance latency compared to unguided swarms that retain over 2,700 dimensions. Concurrently, while the prolonged execution of bIRIME reflects a combination of larger feature subsets and higher candidate generation counts, KDS-EL restricts local refinement to a bounded elite pool, harmonizing thorough exploration depth with practical computational thrift.

4.2.7. *Parameter sensitivity and robustness analysis*

To evaluate algorithmic robustness and justify default parameter configurations, a one-factor-at-a-time sensitivity analysis is conducted following standard evaluation protocols in evolutionary machine learning [36]. Four representative benchmarks across distinct dimensional scales are evaluated: *Adenoma*, *Colon*, *ALL4*, and *BASEHOCK*. Five key hyperparameters governing core strategies are systematically investigated: knowledge weight decay $\gamma \in (0, 1)$, phase split ratio $\kappa \in (0, 1)$, elite perturbation fraction $\phi \in (0, 1)$, mutual information initialization ratio $\rho \in (0, 1)$, and elite pool size $k_e \in \mathbb{N}^+$, with remaining parameters following Table 3. Table 9 summarizes tested ranges, recommended configurations, stable operating bands, and maximum accuracy deviations across the four benchmarks. Furthermore, Fig. 10 illustrates sensitivity profiles and stable operating bands across all parameters, with full evaluation grids in Supplementary Table S6.

Table 9: Parameter sensitivity analysis across representative benchmark datasets.

Parameter	Tested range	Selected	Stable range	Max ΔAcc (pp)
Weight decay	0.90–1.00	0.99	[0.95, 0.995]	±1.2
Phase split	0.3–0.9	0.70	[0.60, 0.80]	±0.7
Flip fraction	0.005–0.10	0.01	[0.005, 0.02]	±1.3
MI init ratio	0.0–1.0	0.50	[0.25, 0.75]	±1.2
Elite count	1–10	3	[2, 5]	±0.9

The empirical sensitivity trajectories reveal that KDS-EL maintains stable optimization dynamics through coordinated phase scheduling and prior guid-

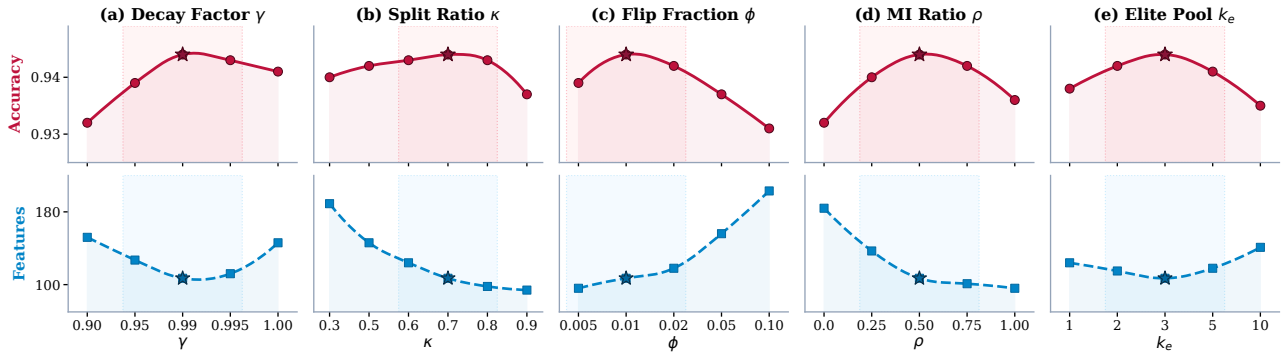


Figure 10: Parameter sensitivity and stability profiles of KDS-EL across investigated hyperparameters.

ance. The phase split ratio κ governs the temporal transition between dimensional pruning in Phase 1 and classification refinement in Phase 2 via Eq. (26). Allocating an insufficient budget to Phase 1 with $\kappa < 0.50$ leaves the swarm encumbered by redundant dimensions; conversely, an overly prolonged Phase 1 with $\kappa > 0.85$ curtails the exploitation horizon necessary for uncovering subtle synergies. The stable interval $\kappa \in [0.60, 0.80]$ confirms that default $\kappa = 0.70$ balances dimensional contraction and precision. Concurrently, the mutual information initialization ratio ρ regulates the trade-off between prior knowledge and exploratory coverage in Eq. (14). Omitting priors when $\rho \to 0$ forces unguided exploration, whereas excessive seeding with $\rho \to 1.0$ homogenizes candidate masks and provokes premature entrapment. The operating band $\rho \in [0.25, 0.75]$ with default $\rho = 0.50$ robustly sustains population diversity while seeding favorable early momentum.

Regarding memory accumulation and local exploitation, sensitivity trajectories confirm that closed-loop learning is robust across wide parameter ranges. The weight decay factor γ in Eq. (20) governs historical retention in $\mathbf{W}$. Aggressive decay with $\gamma < 0.95$ induces premature search amnesia, while unattenuated accumulation at $\gamma = 1.00$ causes obsolete feedback to persist indefinitely. The default setting $\gamma = 0.99$, situated within the stable band $[0.95, 0.995]$, effectively harmonizes knowledge retention with adaptive sensitivity to emergent synergies. Furthermore, perturbation fraction ϕ and pool size k_e calibrate local search within the KER strategy. Setting ϕ excessively large introduces disruptive alterations that fail the greedy pairwise in-place replacement rule in Eq. (25); conversely, the default setting $\phi = 0.01$ within the stable band $[0.005, 0.02]$ provides controlled perturbation that maximizes improving substitutions. Bounding elite pool size to $k_e = 3$ within the stable interval $[2, 5]$ concentrates computation on high-performing exemplars while limiting generational overhead to $k_e \ll N$.

Synthesizing across all five hyperparameters, performance trajectories in Fig. 10 consistently exhibit wide, forgiving basins of attraction. Across identified stable operating bands, maximum accuracy variation is confined within 1.3 percentage points, and KDS-EL maintains stable performance profiles throughout the tested spectrum, indicating favorable parametric tolerance across representative dimensional regimes.

4.2.8. Discussion

A critical empirical phenomenon in high-dimensional feature selection is the accuracy ceiling effect: across eight benchmark datasets, comprising seven biomedical microarrays alongside *COIL20*, classification accuracy exceeds 0.995 with inter-algorithm differences narrowing within 0.003. Consequently, classification accuracy ceases to provide sufficient discriminative resolution, making active feature cardinality paramount for assessing model parsimony [2, 3].

Under this saturated regime, KDS-EL exhibits marked compactness advantages, isolating ultra-sparse candidate feature subsets of 39 to 84 dimensions across these eight ceiling benchmarks, spanning 39 to 83 dimensions across the seven biomedical microarrays alongside 84 dimensions on *COIL20*. In stark contrast, competing metaheuristics and evolutionary filters retain hundreds to thousands of redundant dimensions. This divergence highlights a pervasive failure mode: lacking early-stage dimensional penalization and elite refinement, optimizers succumb to feature bloat once high accuracy is attained. From a translational perspective, contracting candidate subsets to 39 to 83 informative genes on biomedical microarrays substantially scales down the hypothesis space into a tractable candidate regime for downstream experimental validation, whereas uncurated subsets spanning hundreds to thousands of dimensions remain prohibitive for targeted screening.

Beyond nearest-neighbor metrics, a fundamental methodological consideration is inductive bias coupling [4]: whether feature subsets optimized under a 3-NN wrapper overfit its localized metric geometry. While primary evaluations follow the standardized 3-NN wrapper protocol, Supplementary Table S9 reports an exploratory transferability appraisal across Support Vector Machines and Random Forest classifiers on representative benchmarks. Discovered compact subsets provide preliminary empirical evidence of cross-model compatibility alongside localized inductive boundaries. On *Adenoma* and the 12,625-dimensional *ALL4* benchmark, contracting features by over 99.0% elevates classification accuracy across all three learning paradigms, with all classifiers attaining 100% accuracy on *Adenoma* under merely 66 selected dimensions. Conversely, on *Colon*, the ultra-compact 14-feature signature elevates 3-NN accuracy by 9.36 percentage points from 79.23% to 88.59%, while exhibiting localized performance attenuation on Support Vector Machines from 85.51% to 80.51% and on Random Forest from 85.38% to 82.18%. This localized trade-off transparently delineates the inductive boundary: extreme dimensional compression optimized specifically for non-parametric local metric geometry can exhibit modest sensitivity when transferred across divergent hypotheses, such as margin-maximizing hyperplanes or orthogonal axis partitions on small cohorts. Rather than indicating degenerate over-fitting, these exploratory observations indicate favorable cross-paradigm compatibility while delineating the boundary conditions of cross-model feature transfer.

Furthermore, biological annotation of selected dimensions is conducted strictly as an exploratory investigation providing retrospective literature concordance rather than prospective clinical validation. Annotation of prominent candidate genes on representative microarrays reveals clear consistency with characterized oncological mechanisms, as detailed in Supplementary Table S10. In *Colon* adenocarcinoma, identified dimensions show concordance with known molecular alterations: the selected subset includes *COL11A1*, a documented marker of desmoplastic stroma and invasive carcinoma [44]; *GUCA2A*, a mucosal differentiation hormone whose downregulation marks early colorectal neoplasia [45]; alongside smooth muscle structural genes *DES* and *ACTG2*. Similarly, in *Prostate* carcinoma, recovered candidate genes include *HPN* and *AMACR*, widely documented markers in prostate pathology literature [46, 47]. This alignment serves as supportive retrospective consistency evidence indicating that KDS-EL isolates dimensions enriched with biological relevance rather than computational artifacts.

5. Conclusion

This paper introduces KDS-EL, an integrated knowledge-guided dual-swarm evolutionary learning framework for ultra-high-dimensional wrapper feature selection. By orchestrating mutual information soft seeding with adaptive stagnation recovery, differential importance memory with exponential forgetting, and stage-aware dynamic two-phase objective scheduling, KDS-EL establishes an effective closed-loop optimization system. Comprehensive empirical evaluations across 20 high-dimensional benchmark datasets corroborate its competitive multi-criteria performance: attaining 0.9545 mean internal cross-validation accuracy, compressing active feature subsets to merely 87.8 dimensions, and achieving a 3.4-fold acceleration over canonical dual-swarm search. Exploratory evaluations further provide preliminary evidence of cross-model inductive compatibility alongside retrospective biological plausibility across representative benchmarks, while delineating localized trade-offs under divergent classifier inductive biases.

Promising trajectories for future exploration include integrating self-supervised graph embeddings to capture epistatic interactions, establishing

cross-cohort importance transfer mechanisms, extending the knowledge-guided framework toward explicit multi-objective Pareto optimization, and incorporating outer nested cross-validation protocols for prospective clinical translation.

CRediT authorship contribution statement

Author contribution details are documented in the separate Title Page in accordance with the double-blind review policy.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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